

3 Brill in North Sea, Skagerrak, Kattegat, and English Channel

bll.27.3a47de – *Scophthalmus rhombus* in Subarea 4 and divisions 3.a and 7.d–e

Brill (*Scophthalmus rhombus*) is assessed in the Working Group on the Assessment of Demersal Stocks in the North Sea and Skagerrak (WGNSSK) since 2013. Because only official landings and survey data were available, brill in Subarea 27.4, divisions 27.3.a, 27.7.d, e was defined as a category 3 stock and advice was given using the 2 over 3 rule applied to a Dutch commercial beam trawl LPUE index (vessels > 221 kW) (ICES, 2024). In 2022, the WKLIFE X methods were applied and advice was given using the *chr* rule (ICES, 2020). The stock was benchmarked in 2023 where a SPiCT assessment was thoroughly tested and accepted to provide stock status (ICES, 2023a). This assessment, with minor updates, was used in 2024 and 2025 to provide catch advice according to the MSY approach (Category 2 stock, ICES, 2023b). Since 2020, annual advice is requested instead of biennial advice.

3.1 General

3.1.1 Stock definition

The genetic structure of brill over its entire distribution area was characterized by Vandamme (2014). Genetic variation was found to be of intermediate to high levels, but the results showed almost no differentiation between potential biological populations and/or management units. Therefore, brill in 3.a, 4 and 7.d, e is still treated as a single stock that could have an even wider geographical spread. More information can be found in the Stock Annex.

3.1.2 Biology and ecosystem aspects

A general description of the available information on the biology and ecosystem aspects can be found in the Stock Annex.

3.1.3 Fisheries

Brill is mainly a high value bycatch species in fisheries for plaice and sole. Nine countries are involved in the fisheries: Belgium, Denmark, France, Germany, Ireland, The Netherlands, Norway, Sweden and UK (England, Northern Ireland, Scotland and the Channel Islands). In 2024, The Netherlands landed most brill (29.2%), followed by the UK (England) (22.3%) and France (16.9%). Most brill is landed by the TBB fleet (48.1%), followed by fleets using other active gears, such as OTB, SDN and SSC (34.3%) and passive gears, such as GTR and GNS (13.4%).

3.1.3.1 Management

No explicit management objectives have been defined for the brill stock in 3.a, 4, 7.d, e, and no specific management objectives or plans are known to ICES. As a primarily bycatch species, regulations related to effort restrictions for the most important fleets catching brill (e.g. beam trawlers) are likely to impact the stock. Fishing effort has been restricted in the past for demersal fleets

in a number of EC regulations (e.g. EC Council Regulation Nos. 2056/2001, 51/2006, 41/2007, and 40/2008). The most recent one includes the ban of pulse trawling from 1 July 2021 onwards (EC Council Regulation No. 2019/1241). A combined EU TAC for turbot and brill was set in areas 2.a and 4 and applied to EU fisheries until 2023 (Table 3.1.1). In 2024, the combined EU TAC for turbot and brill has been effectively disaggregated, determining specific TAC by areas and species (Table 3.1.2) (EU, 2024).

Table 3.1.1. Historical overview of combined TACs for brill (*Scophthalmus rhombus*) and turbot (*Scophthalmus maximus*) in Division 27.2.a and Subarea 27.4.

Year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
TAC	9000	9000	6750	5738	4877	4550	4323	4323	5263	5263	5263	4642

Year	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
TAC	4642	4642	4642	4642	4488	5924*	7102	8122	6498	5848	5487	3747

* The TAC was increased from 4937 to 5924 at the end of 2017.

Table 3.1.2. Overview of the TAC (in tonnes) for brill (*Scophthalmus rhombus*) in Divisions 27.2.a, 27.3.a, 27.7.d-e and Subarea 27.4. The total represents the sum of all area specific TAC (EU, 2024) including projected third country catches (0.55%) (EU-UK Joint Committee, 2023).

Year	2024	2025
Total	2456	2970

The combined TAC for brill and turbot has been restrictive in 2007, 2015 and 2016 (average overshoot 310 tonnes; Figure 3.1.3.1). In 2016, some of the Member States with a share in the TAC, such as Belgium, Germany and The Netherlands, asked for an advance of their quota for 2017, in order to prevent further overshooting ($\pm 10\%$). The TAC in 2017 was 4937 tonnes, but at the end of the year, it was increased to 5924 tonnes ($\pm 20\%$; 10% to compensate for the advance from 2016 and 10% for 2017). There were several reasons to justify this increase: a) after the inter-benchmark of turbot, a new advice (for 2018) was given, which meant a 148% increase against the previous TAC (2017)¹, b) similar to 2016, member states were asking an advance of their quota for next year (2018), c) observations and catches of fishers did not seem to confirm the assessment (delay with data). Although no new advice was given in 2018 (no reopening), the TAC for 2019 was increased to 8122 tonnes. The reason for this remains unclear. The combined TAC for brill and turbot was not restrictive for the most recent period (2017–2024).

¹ At WGNSSK 2018, a mistake was discovered in the final inter-benchmark run of turbot. This involved an even higher increase.

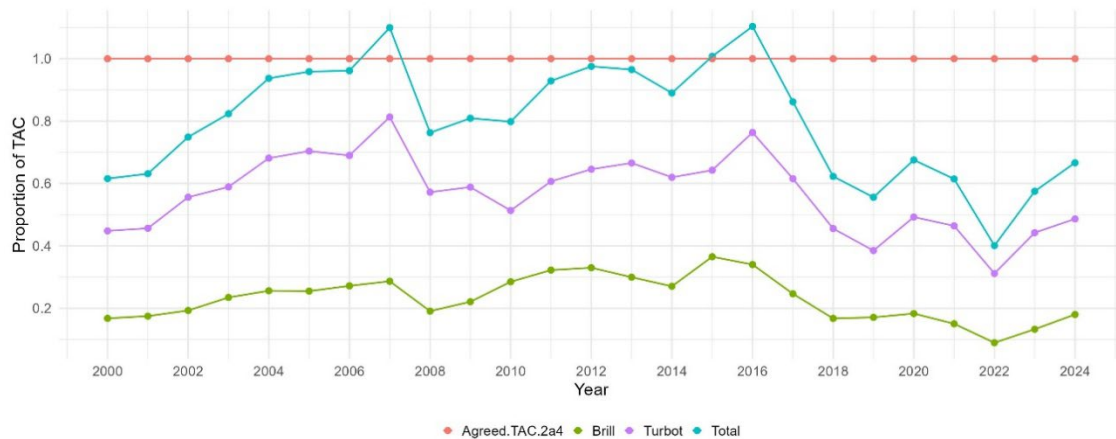


Figure 3.1.3.1. Brill in Subarea 4 and divisions 3.a and 7.d-e. TAC uptake for both brill and turbot in area 2.a and 4.

The introduction of specific TAC for brill (and turbot) in 2024 supports a sustainable management of this stock. Although the inclusion of Division 27.2a in the TAC for brill continues to be a mismatch between management area and the stock area defined by ICES, catches of brill in this Division have historically been very low and are therefore unlikely to significantly affect TAC uptake or the sustainable management of the stock. In 2024, the brill specific TAC came into action and was undershot by 50.75% (1210 landed tonnes) (Figure 3.1.3.2.).

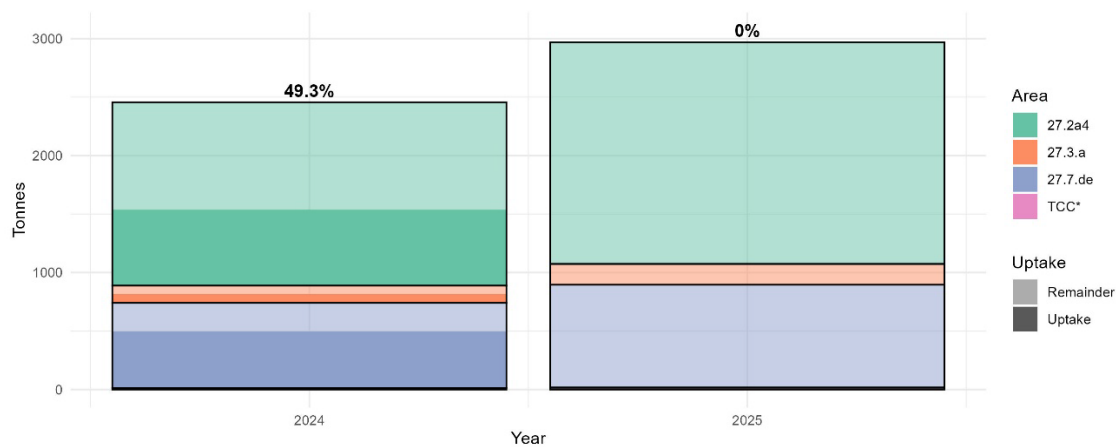


Figure 3.1.3.2. Brill in Subarea 4 and divisions 3.a and 7.d-e. TAC uptake for brill and turbot in divisions 3.a, 2.a, 7.d-e and Subarea 4. No information on the official landings for 2025 is currently available and therefore the TAC uptake for this year cannot be calculated yet.

No restriction on the minimum landing size for brill is imposed by the EC. Some authorities or producer organizations have however installed Minimum Conservation Reference Sizes (MCRS) for brill. Dutch producer organizations increased the MCRS when the TAC was limiting (e.g., from 27 cm to 30 cm in 2016 and later even to 32 cm). Moreover, weekly landings of turbot and brill are often capped to stay within the TAC (especially when the TAC is limiting). Following increases in advice in 2018–2019, PO measures were relaxed. An overview is shown in the Table 3.1.3

Table 3.1.3. Overview of Dutch PO measures for brill.

Dutch PO-measures			
Year	Date	Max kg per week/trip	MCRS
2016	January - March	-	27 cm
2016	April – May	-	30 cm
2016	May – September	-	32 cm
2016	October – November	375 kg	32 cm
2016	November – December	600 kg	32 cm
2017	January – February	-	32 cm
2017	March – October	800 kg	32 cm
2017	November - December	2000 kg	30 cm
2018	January – August	2000 kg	30 cm
2018	September - October	2500 kg	30 cm
2018	October - December	3000 kg	27 cm
2019	January – December	3000 kg	27 cm
2020	January - December	3000 kg	27 cm
2021	January - December	3000 kg	27 cm
2022	January - December	3000 kg	27 cm
2023	January - December	3000 kg	27 cm
2024	January - December	3000 kg	27 cm
2025	January - December		25 cm

Since January 1st 2019, brill is fully under the landing obligation. In 2025, Dutch producer organizations reduced the MCRS from 27 to 25 cm to reduce international inequality with vessels of a different flag not restricted by this measure and to avoid difference in enforcement of this measure in Dutch auctions. There is no more cap on weekly landings for both turbot and brill to 3000 kg as the quotas are now species specific and are not restrictive anymore.. Belgium applied a MCRS of 30 cm from January 1st 2017. However, this was raised to 32 cm from 23/12/2020 onwards.

3.1.3.2 ICES advice for 2024

The ICES advice for 2024 was:

ICES advises that when the MSY approach is applied, catches in 2024 should be no more than 2456 tonnes.

The stock status was presented as follows:

		Fishing pressure				Stock size		
		2020	2021	2022		2021	2022	2023
Maximum sustainable yield	F _{MSY}	✓	✓	✓ Below		MSY B _{trigger}	✓	✓ Above
Precautionary approach	F _{pa} , F _{lim}	✓	✓	✓ Below possible reference points		B _{pa} , B _{lim}	✓	✓ Above possible reference points
Management plan	F _{MGT}	—	—	— Not applicable		B _{MGT}	—	— Not applicable

3.1.3.3 ICES advice for 2025

The ICES advice for 2025 was:

ICES advises that when the MSY approach is applied, catches in 2025 should be no more than 2970 tonnes.

The stock status was presented as follows:

		Fishing pressure				Stock size		
		2021	2022	2023		2022	2023	2024
Maximum sustainable yield	F _{MSY}	✓	✓	✓ Below		MSY B _{trigger}	✓	✓ Above trigger
Precautionary approach	F _{pa} , F _{lim}	?	?	? Undefined		B _{pa} , B _{lim}	?	? Undefined
Management plan	F _{MGT}	—	—	— Not applicable		B _{MGT}	—	— Not applicable

3.2 Data

From 2015 onwards, also discards by métier were requested from all countries contributing to the fishery of this stock through InterCatch. For the WGNSSK data call in 2017, all available age and length data were requested through InterCatch for three years back in time (2014–2016). For the WGNSSK data calls from 2018 onwards, similarly both age and length data were requested from discards and landings.

3.2.1 Landings

3.2.1.1 Official landings

Supplementary Table 3.3.2.1, Supplementary Table 3.3.2.2 and Supplementary Table 3.3.2.3 summarize the official brill landings by country for Division 3.a, Subarea 27.4, and divisions 27.7.d-e respectively (Source: ICES FishStat). The total official landings by area can be consulted in Table 3.2.1. and Figure 3.2.1. Over the period 1950–1970, total landings remained under 1000 tonnes (range: = 611–987 tonnes), followed by a gradual increase to 1564 tonnes in 1977. From 1978 onwards, total landings remained higher than 1500 tonnes (range: 1538–3141 tonnes). In 1993, a maximum of 3141 tonnes was caught. From 2010–2020, total annual landings fluctuated around an average of 2211 tonnes (range = 1903–2537 tonnes). In 2015, landings peaked at 2537 tonnes to then continuously decrease until 2023, reaching levels comparable to 1971, with only 1073 landed tonnes in 2023. Landings appear to remain more or less constant in the last three years with a small increase in 2024: 1210 tonnes.

Table 3.2.1. Brill in Subarea 4 and divisions 3.a and 7.d-e. Total official landings (tonnes) of brill in the 27.3a47de (Greater North Sea) over the period 1950–present, subdivided into Subarea 27.4 and Divisions 27.3.a and 27.7.d, e (Source: ICES Fishstat). *Preliminary.

Year	3.a	4	7.de	TOTAL
1950	319	384	59	762

Year	3.a	4	7.de	TOTAL
1951	337	511	78	926
1952	236	565	72	873
1953	246	589	62	897
1954	234	529	60	823
1955	212	571	61	844
1956	213	516	60	789
1957	148	468	70	686
1958	203	480	67	750
1959	233	424	59	716
1960	318	486	52	856
1961	305	581	47	933
1962	207	591	55	853
1963	120	620	51	791
1964	106	565	60	731
1965	155	535	48	738
1966	187	546	53	786
1967	106	409	67	582
1968	100	579	57	736
1969	99	658	190	947
1970	97	618	59	774
1971	104	1073	66	1243
1972	120	994	75	1189
1973	131	989	90	1210
1974	200	1152	81	1433
1975	187	1222	137	1546
1976	224	1208	285	1717
1977	388	1410	323	2121
1978	216	1375	411	2002
1979	184	1363	459	2006

Year	3.a	4	7.de	TOTAL
1980	82	1033	402	1517
1981	59	1218	490	1767
1982	74	1294	487	1855
1983	83	1448	526	2057
1984	97	1522	531	2150
1985	110	1709	494	2313
1986	107	1207	456	1770
1987	105	970	493	1568
1988	101	1085	452	1638
1989	97	1302	425	1824
1990	128	893	543	1564
1991	99	1682	470	2251
1992	146	1810	463	2419
1993	212	2439	490	3141
1994	222	1916	490	2628
1995	151	1434	558	2143
1996	112	1247	608	1967
1997	106	957	501	1564
1998	132	1283	451	1866
1999	157	1280	240	1677
2000	142	1508	678	2327
2001	98	1573	738	2409
2002	89	1302	716	2108
2003	129	1346	759	2233
2004	156	1249	666	2071
2005	133	1160	611	1904
2006	140	1175	648	1963
2007	160	1239	743	2142
2008	181	1004	595	1781

Year	3.a	4	7.de	TOTAL
2009	146	1162	594	1902
2010	122	1499	700	2321
2011	131	1496	665	2292
2012	120	1532	624	2276
2013	92	1390	605	2088
2014	79	1255	645	1978
2015	145	1695	698	2537
2016	168	1526	721	2415
2017	170	1460	663	2292
2018	125	1188	714	2027
2019	139	1387	660	2186
2020	162	1189	551	1903
2021	143	879	549	1571
2022	76	490	512	1078
2023*	78	497	497	1073
2024*	78	649	484	1210

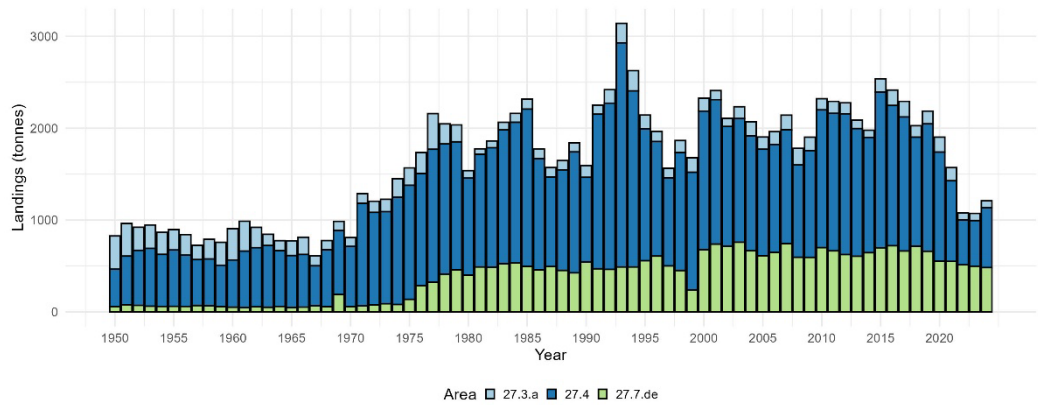


Figure 3.2.1. Brill in Subarea 4 and divisions 3.a and 7.d-e. Official landings (tonnes) over the period 1950–present per area, as officially reported (Rec 12; ICES Fishstat).

Subarea 27.4 accounts for a major part of the landings (Figure 3.2.2), on average generating $66\pm 8\%$ of the total landings over the time-series (range: 45.4–86.7%). The English Channel and the Skagerrak-Kattegat area are responsible for average contributions to the international brill landings of 21.2% and 12% respectively. Skagerrak-Kattegat was responsible for a higher relative importance in the total landings during the first two decades of the time-series, and the English Channel has gained importance since the late seventies. In 2022, and for the first time in the entire

time series, the English Channel (Divisions 27.7.d-e) accounted for the major part of the landings, with 47.7% of the total landings, due to a persistent decrease of landings from Subarea 27.4. In 2024, the relative proportion of landings was 53.6%, 40% and 6.4% for Subarea 27.4, Division 27.7.de and 27.3a respectively (Table 3.2.2).

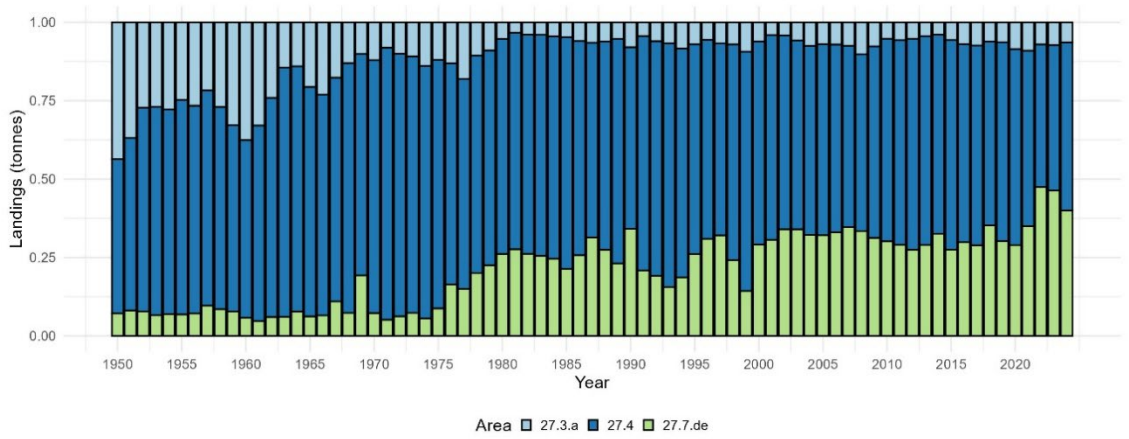


Figure 3.2.2. Brill in Subarea 4 and divisions 3.a and 7.d-e. Relative contribution to the total official landings for brill per area to the total international landings (tonnes) in the Greater North Sea over the period 1950–present (Source: ICES Fishstat).

Table 3.2.2. Brill in Subarea 4 and divisions 3.a and 7.d-e. Overview of absolute official landings per area over the last 15 years with an indication of the relative proportion by Subarea/Division (Source: ICES Fishstat). *Preliminary.

Year	Absolute landings (tonnes)				Relative proportion		
	3a	4	7de	TOTAL	3a	4	7de
2010	122	1499	700	2321	5.2	64.6	30.1
2011	131	1496	665	2292	5.7	65.3	29
2012	120	1532	624	2276	5.3	67.3	27.4
2013	92	1390	605	2088	4.4	66.6	29
2014	79	1255	645	1978	4	63.4	32.6
2015	145	1695	698	2537	5.7	66.8	27.5
2016	168	1526	721	2415	6.9	63.2	29.9
2017	170	1460	663	2292	7.4	63.7	28.9
2018	125	1188	714	2027	6.2	58.6	35.2
2019	139	1387	660	2186	6.3	63.5	30.2
2020	162	1189	551	1903	8.5	62.5	29
2021	143	879	549	1571	9.1	55.9	35
2022	76	490	512	1078	7.1	45.4	47.5
2023*	78	497	497	1073	7.3	46.3	46.4

Year	Absolute landings (tonnes)				Relative proportion		
	3a	4	7de	TOTAL	3a	4	7de
2024*	78	649	484	1210	6.4	53.6	40

Over the period 1950–1970, landings were dominated by the UK and Denmark. From the 70’s onwards, the Netherlands dominated the landings of the stock, but faced a steady and rapid decrease in the last 3 years. The proportion of landings coming from the UK and the Netherlands are similar in 2024 (Figure 3.2.3 and Figure 3.2.4).

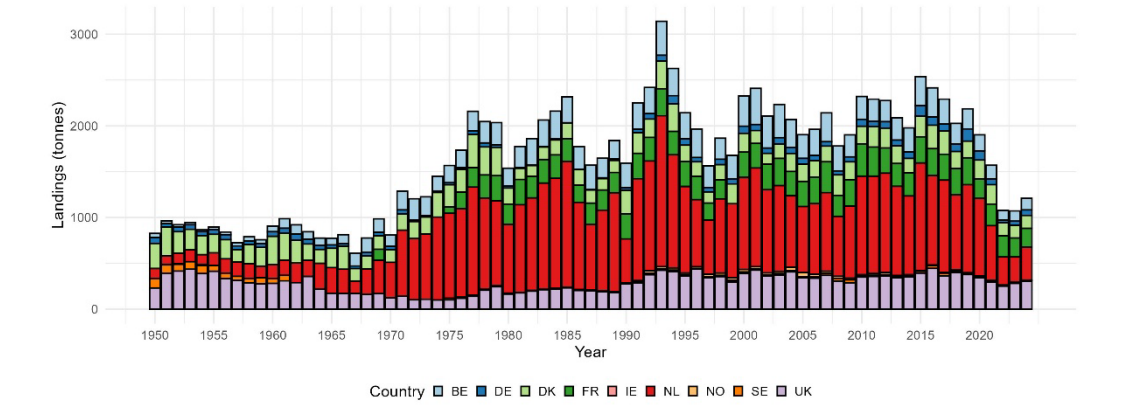


Figure 3.2.3. Brill in Subarea 4 and divisions 3.a and 7.d-e. Official landings (tonnes) over the period 1950–present per country, as officially reported (Rec 12; ICES Fishstat).

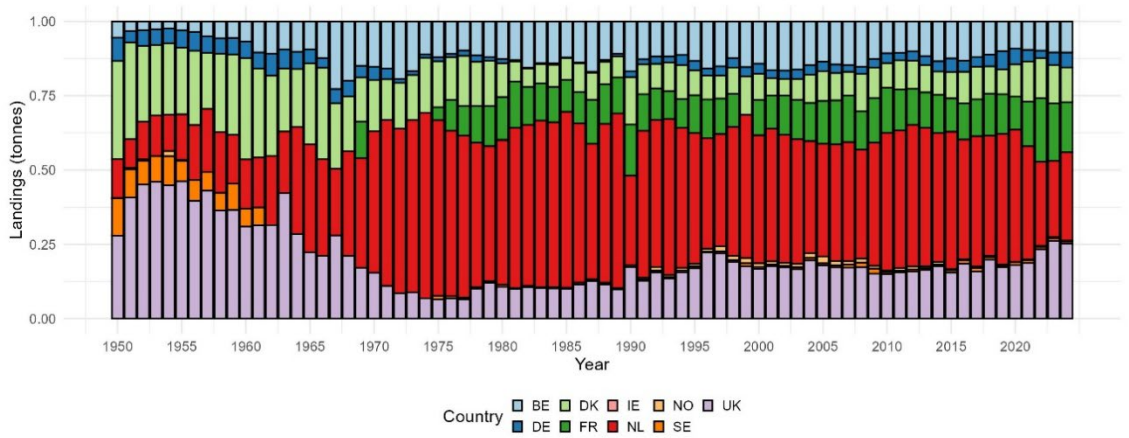


Figure 3.2.4. Brill in Subarea 4 and divisions 3.a and 7.d-e. Relative contribution to the official landings for brill per country to the total international landings (tonnes) in the Greater North Sea over the period 1950–present (Source: ICES Fishstat).

3.2.1.2 InterCatch

From 2014 onwards, data are available in InterCatch. Table 3.2.3 and Table 3.2.4. and Figure 3.2.5 and Figure 3.2.6, shows the ICES landings estimates (both discards and landings provided through InterCatch). Figure 3.2.7 compares the InterCatch landings estimates with the official landings by country for 2024. The Netherlands fished the majority of the catches (predominantly in Subarea 4), followed by the UK and France. Denmark was responsible for the majority of the landings in Division 27.3a. The UK and France had the highest landings in Division 27.7d and 27.7e (Table 3.2.5). The most important gear types landing brill are TBB and OTB, followed by GTR (Table 3.2.6).

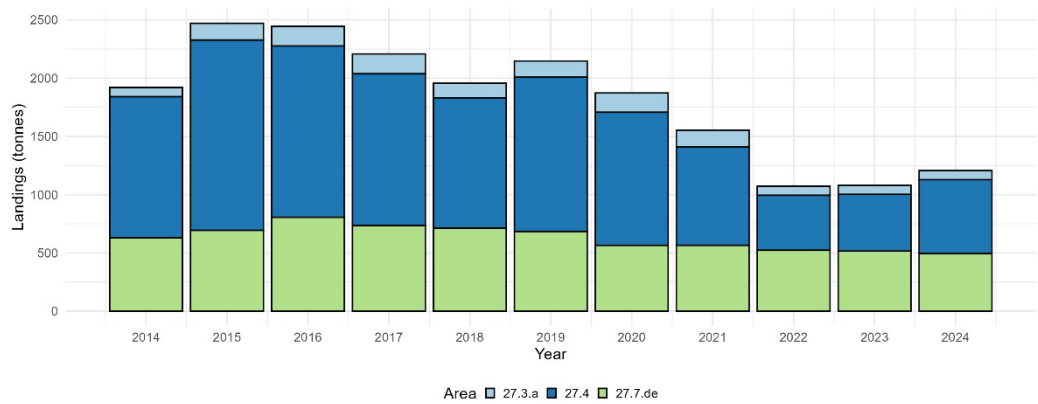


Figure 3.2.5. Brill in Subarea 4 and divisions 3.a and 7.d-e. ICES landings estimates (InterCatch, IC) over the period 2014–present by area.

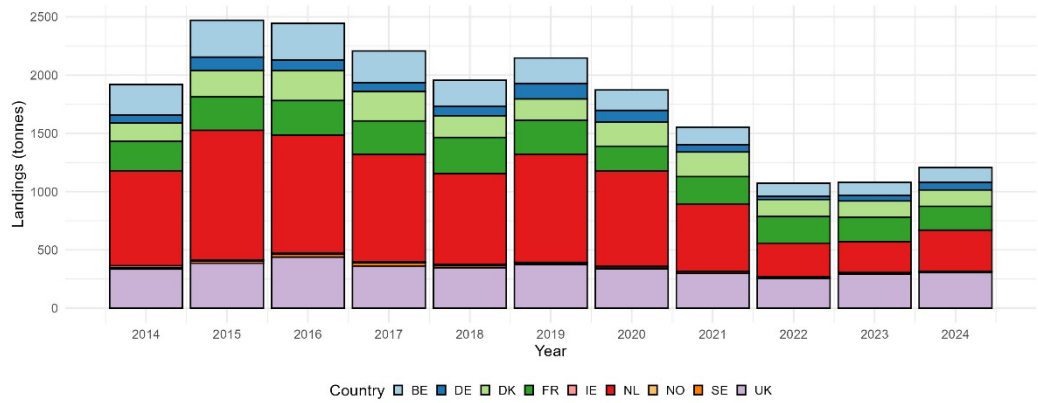


Figure 3.2.6. Brill in Subarea 4 and divisions 3.a and 7.d-e. ICES landings estimates (InterCatch, IC) over the period 2014–present by country.

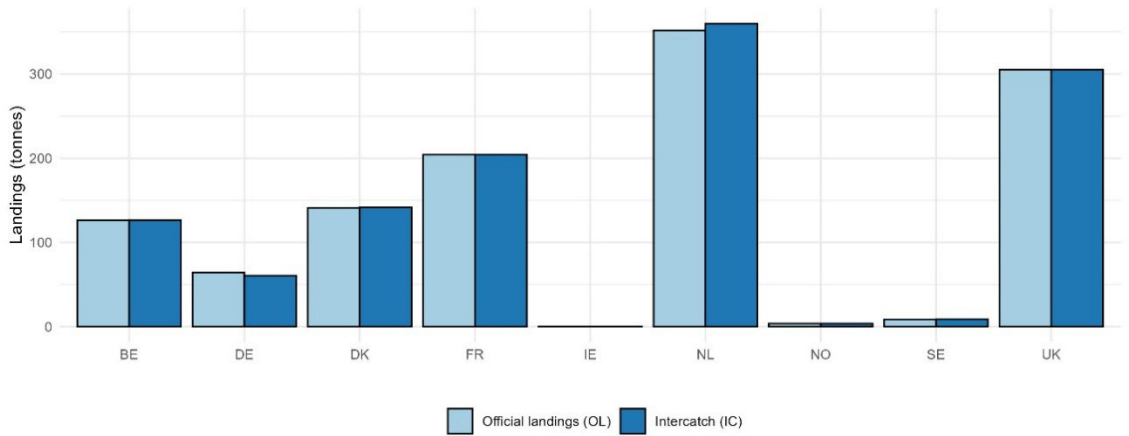


Figure 3.2.7.Brill in Subarea 4 and divisions 3.a and 7.d-e. Comparing ICES landings estimates (InterCatch, IC) to the official landings by country for 2024

Table 3.2.3. Brill in Subarea 4 and divisions 3.a and 7.d-e. Overview of landings (in tonnes) reported to InterCatch (ICES) from 2014-2024.

Year	3.a	4	7.de	TOTAL
2014	79	1210	631	1920
2015	144	1633	693	2470
2016	169	1470	805	2444
2017	170	1302	735	2207
2018	125	1118	713	1956
2019	139	1323	684	2147
2020	166	1142	565	1872
2021	143	844	565	1553
2022	78	471	523	1072
2023	78	487	515	1080
2024	77	634	495	1206

Table 3.2.4. Brill in Subarea 4 and divisions 3.a and 7.d-e. Overview of discards (in tonnes) reported to InterCatch (ICES) from 2014–present.

Year	3.a	4	7.de	TOTAL
2014	49	131	37	217
2015	76	91	40	207
2016	23	143	50	216
2017	47	121	42	210
2018	88	151	105	343
2019	53	281	58	392
2020	71	75	37	182
2021	18	28	16	62
2022	10	14	10	34
2023	6	45	31	82
2024	9	18	12	38

Table 3.2.5. Brill in Subarea 4 and divisions 3.a and 7.d-e. Overview of 2024 catches (in tonnes) reported to InterCatch (ICES) by country and Subarea/Division (rounded to the nearest tonne).

COUNTRY	3.a			4			7.d-e			Total			
	LAN	DIS	BMS	LAN	DIS	BMS	LAN	DIS	BMS	LAN	DIS	BMS	ALL
Belgium				51	2		75	3		126	5	0	131
Germany	1	0		63	2		0	0		64	2	0	66
Denmark	57	5	0	84	1	0	0			141	6	0	147
France				0	0		204	7		204	7	0	211
Ireland							0	0		0	0	0	0
The Netherlands	8	0		341	8		2	0		351	8	0	360
Norway	2	0		2	0					4	0	0	4
Sweden	9	3	0	0	0	0				9	3	0	12
UK (ENG)				92	5	0	213	2	0	305	7	0	312
Total	77	9	0	634	18	0	495	12	0	1206	38	0	1244

Table 3.2.6. Brill in Subarea 4 and divisions 3.a and 7.d-e. Overview of 2024 landings (in tonnes) for the most important gear types per Subarea/Division (Source: InterCatch).

Gear type	3.a	4	7.d-e	Total
TBB	8	351	222	581
OTB	62	250	97	409
GTR	1	0	118	119
MIS	1	3	43	47
GNS	5	28	9	42
SSC	0	2	3	5
Others	1	0	4	5
Total	78	634	496	1208

For the WGNSSK data call in 2017, available age and length data were requested through InterCatch for three years back in time (2014–2016). From 2018 onwards, the WGNSSK data call also asked for both age and length. Data quality of age readings has been verified in 2019 by an international otolith exchange coordinated by WGBIOP and appeared successful (ICES, 2019). For assessment purposes, age/length allocations in InterCatch does not need to be performed. However, during WGNSSK 2022, length distributions were utilized to investigate the *chr* rule as basis for advice, but these are not used anymore since the stock was upgraded to Category 2 using a SPiCT model, as a result of the most recent benchmark (ICES, 2023a). Nevertheless, during the WKBMSPiCT2 2023 benchmark, length distributions were analysed to investigate which lengths occurred in commercial catches. This helped in the development of an *exploitable* biomass

index for brill using survey data (ICES, 2023a). Length distributions were allocated per catch category (landings and discards (incl. BMS) separately). Further information can be found under §3.2.2 and in the stock annex.

3.2.2 Discards

Due to its high economical value, brill is not expected to be discarded a lot by fishers as long as the quota have not been fully taken. Since January 2019, the stock is completely under the landing obligation. Although discards are not used directly in the assessment model (only landings), they are used to calculate the discard rate to top up the landing advice to catch advice.

InterCatch raising procedures were reviewed during the WKBMSYSPiCT2 2023 benchmark (ICES, 2023a). If discards were not imported for a particular year-quarter-country-métier combination, they were assumed to be unknown (non-zero) and therefore raised. Discards on a year-quarter-country-métier basis were automatically matched by InterCatch to the corresponding landings. From 2014–2017, manual matching had to be done for some strata of The Netherlands and Scotland because discards estimates were provided on year level, while landings were provided per quarter (or the other way around). The matched discards-landings provided a landing-discard ratio estimate, which was then used for further raising (creating discard amounts) of the unmatched discards. The weighting factor for raising the discards was ‘Landings CATON’ (landings catch). No discard estimates were calculated for the industrial bycatch strata (MIS_MIS_0_0_0_IBC).

For every year, it was investigated whether the proportion of landings for which discards weights were available was **equal or larger than 50%** compared to the total landings of that specific **gear group**. When the threshold was met, discard raising was performed on a **gear group level**. Otherwise, discards were raised using all available strata regardless of gear, season, country or area (“**ALL by ALL**”). For the period 2014–2017 this was the case. For 2018–2024, the 50% threshold was reached for some **gear groups**, hence, discard raising for these strata were performed on a **gear group level**. No raising was performed at a season, area or country level. The main reason being the fact that brill is a data-limited stock as a result of opportunistic sampling of this bycatch species, hence seasonal, area and country specific data on discards is scarce, hampering a more detailed/specific raising scheme. A new data call, could potentially provide additional data on discards across seasons, countries and areas, allowing a revision of the raising scheme.

The following groups were distinguished based on the gear:

- TBB
- OTB, OTT, SSC, SDN
- GTR and GNS
- The remaining gears were combined in a REST group

To raise the discards of the REST group, all available strata were considered. However, discard rates higher than 100% from strata that had already been used in a specific gear group were excluded. For example, in 2024, discard raising at the **gear group level** was performed for the OTB and TBB groups. Three specific strata (*i.e.* 3 OTB fleets) reported a discard rate >100%, and were therefore excluded from the raising of the REST group. Apart from these, all other available strata were included when raising discards from the REST group.

Discard data from 2014–2024 are available in InterCatch. Due to a new upload by the UK and the Netherlands, the raising of the 2021–2023 data was done again during WGNSSK 2025. For the 2021 and 2023 data, only the TBB and OTB gear groups reached the threshold and were raised

separately, while the rest of unsampled strata were raised using all available sampled data. For the 2024 data, the threshold for raising by gear group was reached for TBB, OTB and GTR groups. The proportion of landings for which discard weights are available in 2024 was 56.7%. The proportion of imported discards is 54% in 2024 (Table 3.2.7), which is comparable to recent years. An overview of the proportion of discards provided by area and country can be found in Figure 3.2.8 and Figure 3.2.9 respectively, while an overview of the absolute discards by gear can be found in Figure 3.2.10 (b).

Table 3.2.7. Brill in Subarea 4 and divisions 3.a and 7.d-e. Overview of imported and raised data from InterCatch for 2024

Catch category	Survey	CATON (kg)	Percentage
BMS landing	Imported data	52	100
Discards	Imported data	20796	54
	Raised discards	17614	46
Landings	Imported data	1205722	100
Logbook registered discard	Imported data	0	-
TOTAL		1244184	100

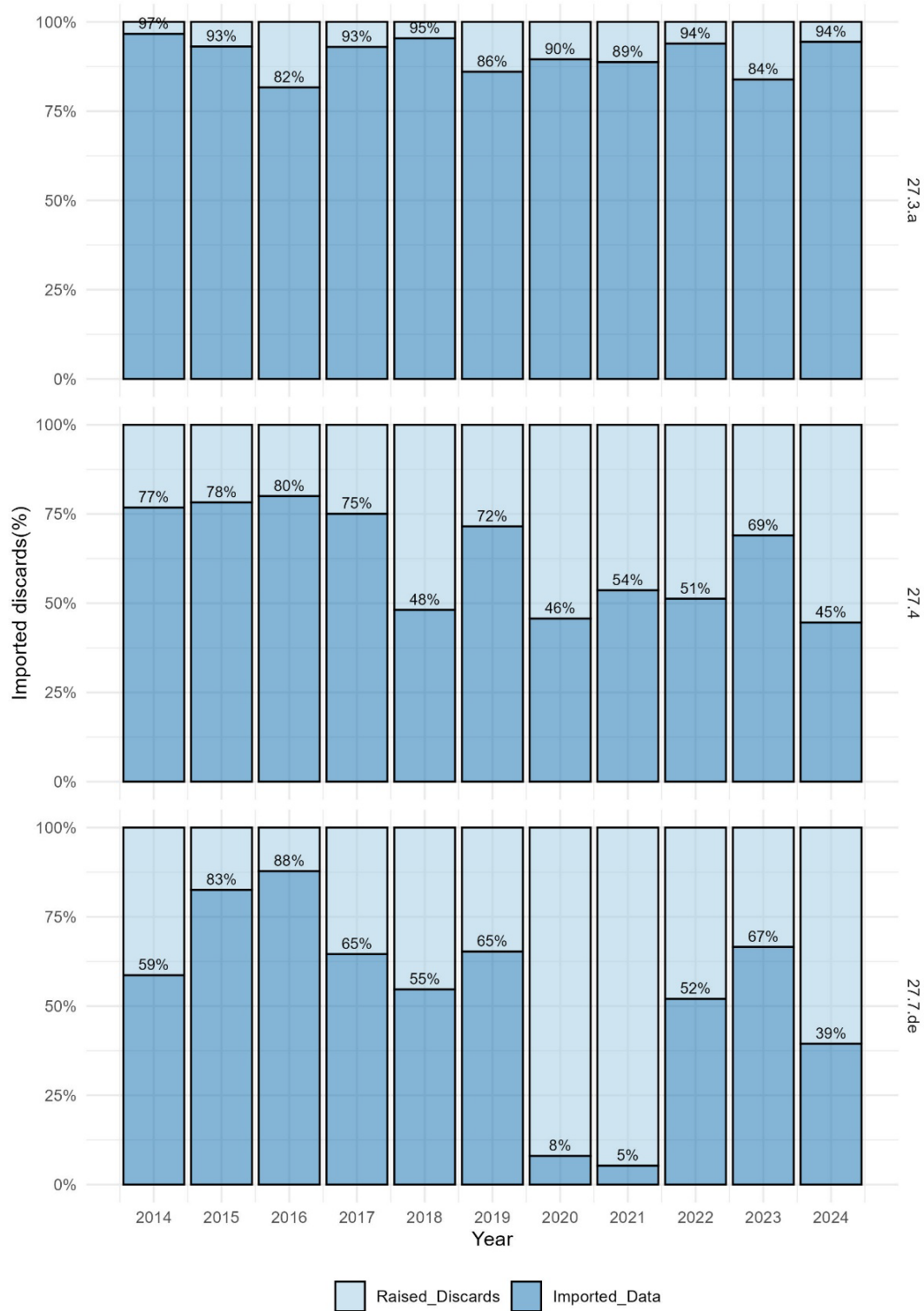


Figure 3.2.8. Brill in Subarea 4 and divisions 3.a and 7.d-e. Proportion of discard weight imported/raised by area and year (2014–present).

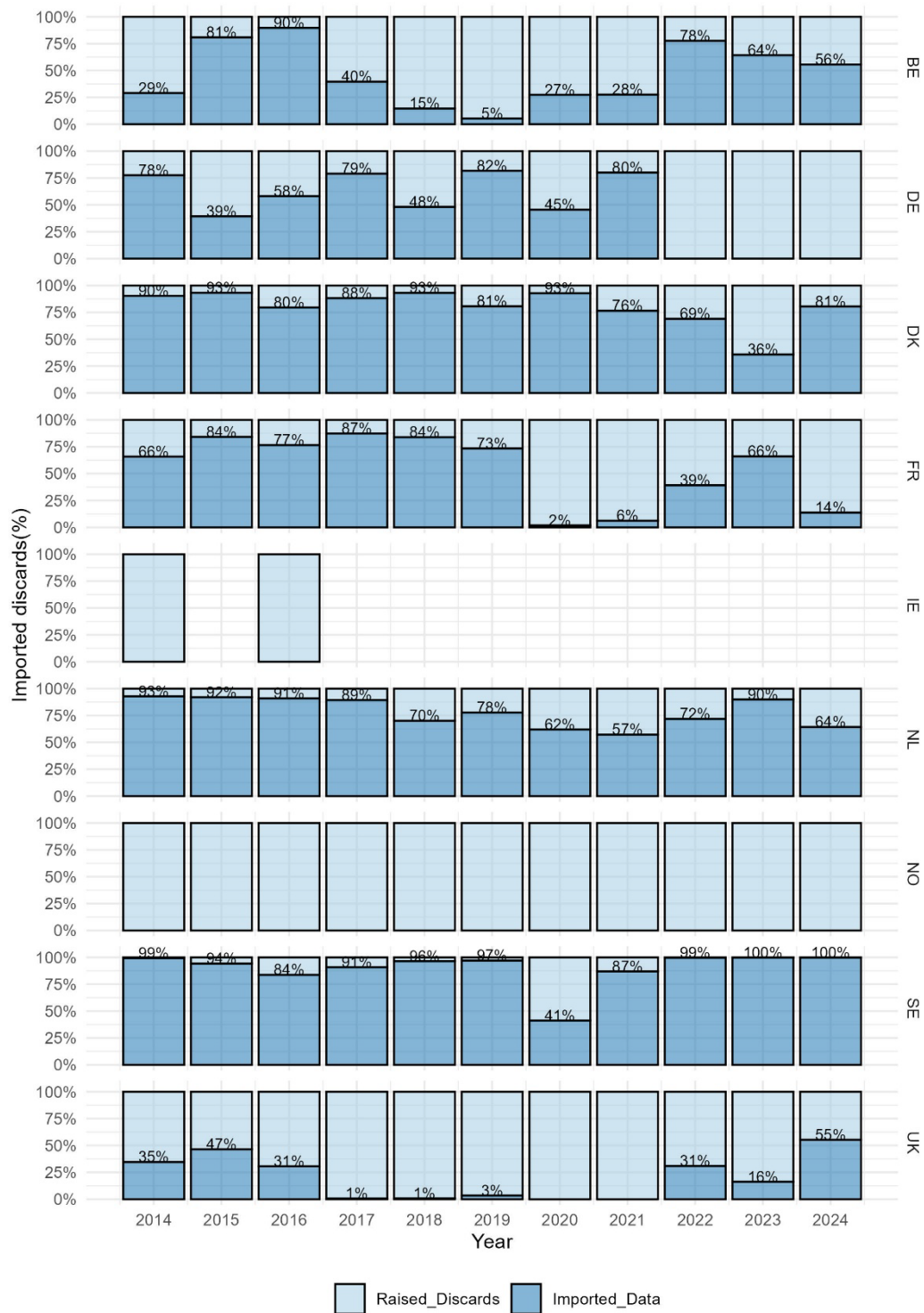


Figure 3.2.9. Brill in Subarea 4 and divisions 3.a and 7.d-e. Proportion of discard weight imported/raised by country and year (2014-present).

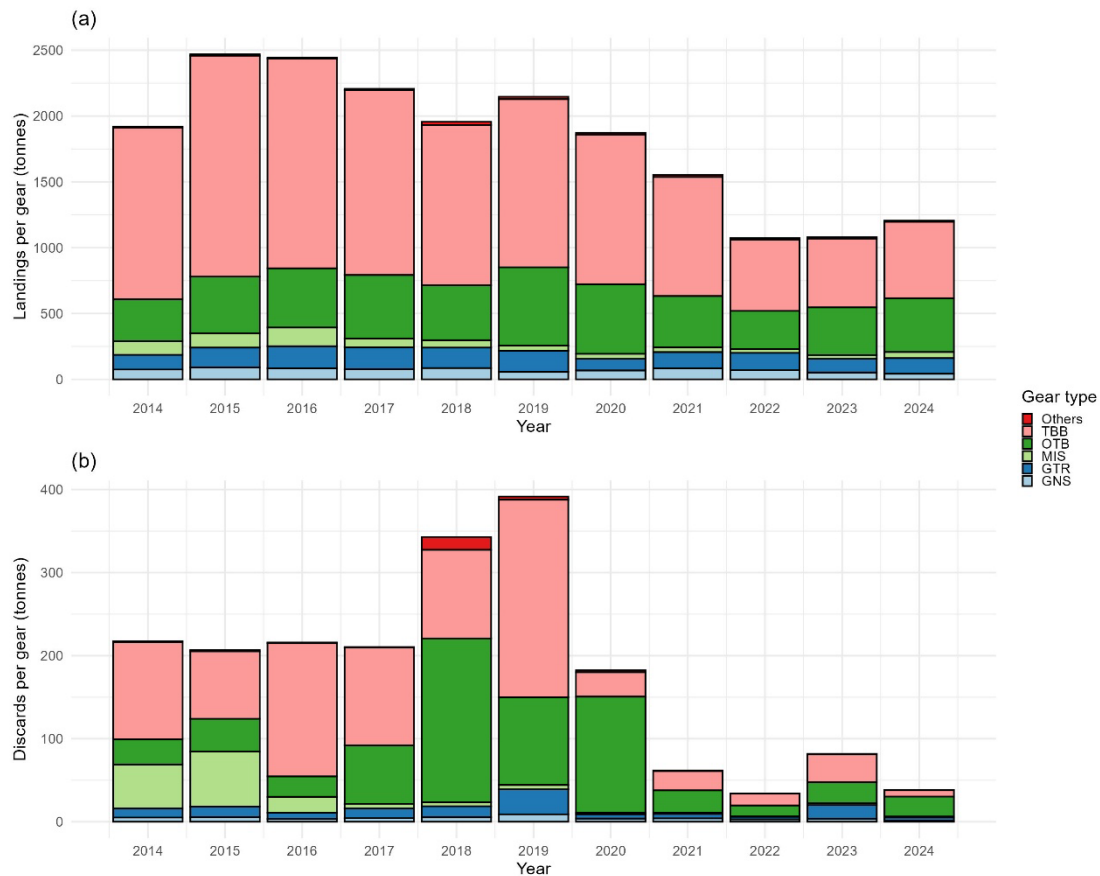


Figure 3.2.10. Brill in Subarea 4 and divisions 3.a and 7.d-e. ICES estimates (InterCatch) of landings (a) and discards (b) per gear type in tonnes (GTR: trammel net, TBB: beam trawl, GNS: gillnet, OTB: otter trawl and MIS: other gears).

An overview of discards and discard rates broken down by country and Subarea/Division in Table 3.2.8 and Table 3.2.9, respectively, while the overall discards and discard rates from 2014–2024 are shown in Table 3.2.10 and for 2022–2024. In 2024, discard rates are only significantly higher than the overall rate (3.09%) for Sweden (25.54%). Discard rates are similar across the distribution of the stock. In 2024, the overall discard rate decreased to 3.09%, but continues to be within the range of observed discard rates for the last 10 years. Brill in the greater North Sea is still considered as a data-limited stock. This means that countries might apply different thresholds to supply their data to InterCatch than for stocks that are not data-limited. For Germany, the exceptionally high discard rate observed in 2019 (41%) was influenced by only one sampled trip having a very high discard rate.

Table 3.2.8. Brill in Subarea 4 and divisions 3.a and 7.d-e. Discard rates (%) for brill by country for 2022–2024 (Source: InterCatch).

Country	2022	2023	2024
Belgium	2.6	5.47	3.75
Denmark	4.57	4.15	4.35
France	2.17	10.09	3.32
Germany	2.59	4.45	3.03

Country	2022	2023	2024
Netherlands	3.19	11.37	2.22
Norway	3.94	6.73	4.38
Sweden	34.31	25.59	25.54
UK England	1.51	2.04	1.52
UK Scotland	1.95	1.36	7.1
Overall	3.08	7.04	3.09

Table 3.2.9. Brill in Subarea 4 and divisions 3.a and 7.d-e. Discard rates (%) for brill by Subarea/Division for 2022–2024 (Source: InterCatch).

Subarea/ Division	2022	2023	2024
27.3.a	11.34	7.61	10.08
27.4	2.88	8.41	2.71
27.7.d-e	1.9	5.62	2.4
Overall	3.08	7.04	3.09

Table 3.2.10. Brill in Subarea 4 and divisions 3.a and 7.d-e. Overall discards and discard rates (all countries and métiers) for brill over the period 2014–present (Source: InterCatch).

Year	Discards	Discard rate (%)
2014	217	10.17
2015	207	7.72
2016	216	8.11
2017	210	8.7
2018	343	14.91
2019	392	15.42
2020	182	8.87
2021	62	3.82
2022	34	3.08
2023	82	7.04
2024	38	3.09

Although not used in the assessment as explained above, length distributions were raised in InterCatch and allocated per catch category (landings and discards (incl. BMS) separately) and by gear groups. The same gear groups (TBB, OTB, GTR and REST) as used for discard raising

were applied. The length distribution allocation of unsampled strata was performed at a **gear group level** when the proportion of landings/discards with sampled length distribution for this **gear group** was found to be higher than 50% and when length distributions were estimated based on a minimum of 100 length measurements. Otherwise, unsampled strata were pooled in the REST group and length distributions were allocated using all available sampled data (“**ALL by ALL**”). For the 2024 data, these thresholds were only reached for the TBB and GTR group for the landings. All remaining unsampled strata were pooled in the REST group and lengths were allocated using all available sampled data. The weighting factor was ‘Mean Weight weighted by numbers-at-length’. Length allocation of industrial bycatch was done using all available strata from landings, discards and BMS landings.

3.2.3 BMS landings

The brill stock is under the landing obligation since January 2019. The official catch statistics have reported BMS landings from 2018 onwards, with 681 kg in 2018, 2036 kg in 2019, 779 kg in 2020, 489 kg in 2021, 220kg in 2022, 744 kg in 2023 and 426kg in 2024. In InterCatch, only 4 kg were reported in 2019 (0 kg prior to 2019), 9 kg in 2020, 124 kg in 2021, 38kg in 2022, 415kg in 2023 and 52kg in 2024. BMS landings are raised together with discards as is described in §3.2.2.

3.2.4 Logbook registered discards

No non-zero logbook registered discards were uploaded to InterCatch.

3.2.5 Indices of biomass

Catches of brill are generally low during surveys. This often results in an underrepresentation of some year or length classes (mainly the older or bigger ones), leading to poorer quality of the resulting survey biomass series and indices, and poor agreement among different surveys. Furthermore, it is important that the survey indices used for this stock are obtained from surveys covering the entire stock area. To remediate and consider these issues, WKBSYSPiCT2 2023 (ICES, 2023a) tested a standardized survey index derived from all available DATRAS survey data in the stock area (Subarea 4 and Divisions 3.a and 7.d–e) following the routine prepared during WKFISHDISH2 (ICES, 2023c). The main objective was to obtain an exploitable biomass index for the entire stock area and move away from spatially-limited indices based on national fleets or single surveys.

The compiled survey dataset was filtered to the stock’s area and included 6 surveys (BITS, BTS, DYFS, FR-CGFS, NS-IBTS and SNS) executed by 8 countries (Belgium, Denmark, France, Germany, Great-Brittain, The Netherlands, Norway and Sweden). The complete survey dataset spanned the period 1971–2024 and 3 quarters (1, 3 and 4), however only data between 1983–present were included for the calculation of the combined biomass survey index. Three different gears were used by 42 different vessels: BT used in BTS, DYFS and SNS (different types of beam trawls); GOV_CL used in FR-CGFS and NS-IBTS (Demersal trawl with clean gear and high headline net) and TV used in BITS (TV-3 trawls). Night hauls were removed to avoid considering different catchability related to different activity regimes during night-time. Weights are calculated for all length samples using an overall (all years included) length-weight key combining all available information in DATRAS. The spatio-temporal distribution of the CPUE (kg/km²) for all surveys combined in the stock area is shown in the Figure 3.2.11. More details on the input data and it’s processing can be found in Annex 2 of the WKBSMYSPICT 2 report (ICES, 2023a).

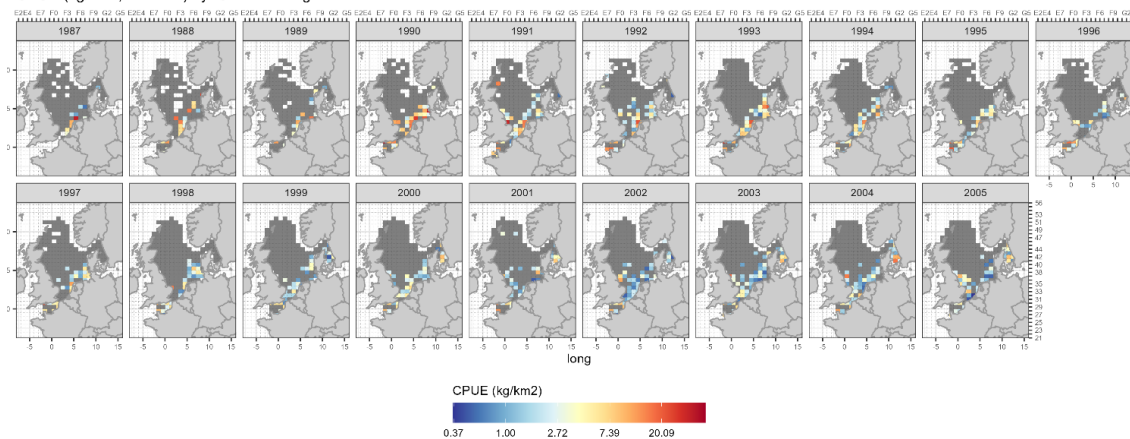
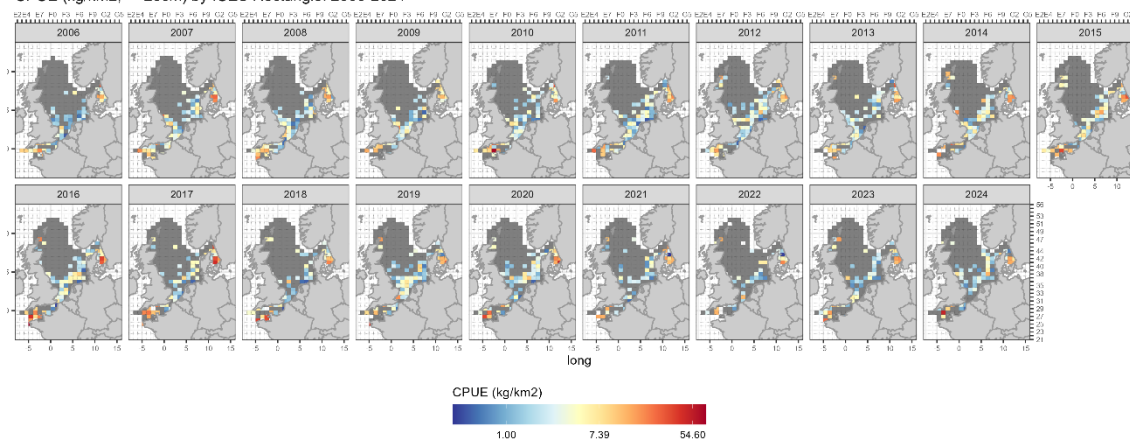
CPUE (kg/km², >=25cm) by ICES Rectangle: 1987-2005CPUE (kg/km², >=25cm) by ICES Rectangle: 2006-2024

Figure 3.2.11. Brill in Subarea 4 and divisions 3.a and 7.d-e. CPUE (kg/km²) for brill >= 25 cm per year and ICES statistical rectangle for all surveys combined in the stock area for 1987–2024.

One GAM model structure was used to estimate 3 indices. As input for the model to calculate the survey index, only brill equal to or larger than 25 cm were retained to allow estimating the exploitable biomass. The model contained a fixed spatial effect (2 dimensional smoother), a temporal trend (1 dimensional smoother), a spatio-temporal effect (3 dimensional smoother), a depth effect (1 dimensional smoother), a fixed gear effect, a random ship effect, a linear offset on the swept-area and the observation error was assumed to follow a Tweedie distribution.

To calculate the first index (Table 3.2.11 and Figure 3.2.12), data were filtered to 1983–1998 and contained only data on quarter 1 and from Division 3a and Subarea 4 (no sampling in English Channel). Only data collected by the GOV_CL gear remained. The gear and seasonal effect were therefore removed from the model. The second and third indices (Table 3.2.12 and Figure 3.2.12) were generated with the GAM model using data filtered to the period 1999–2024 for each semester of the year. This period corresponds to when most of the stock area was sampled and when there was consistent coverage in terms of gears, quarter and area. All indices and their coefficient of variation (standardized to the mean) were included in the final SPiCT assessment.

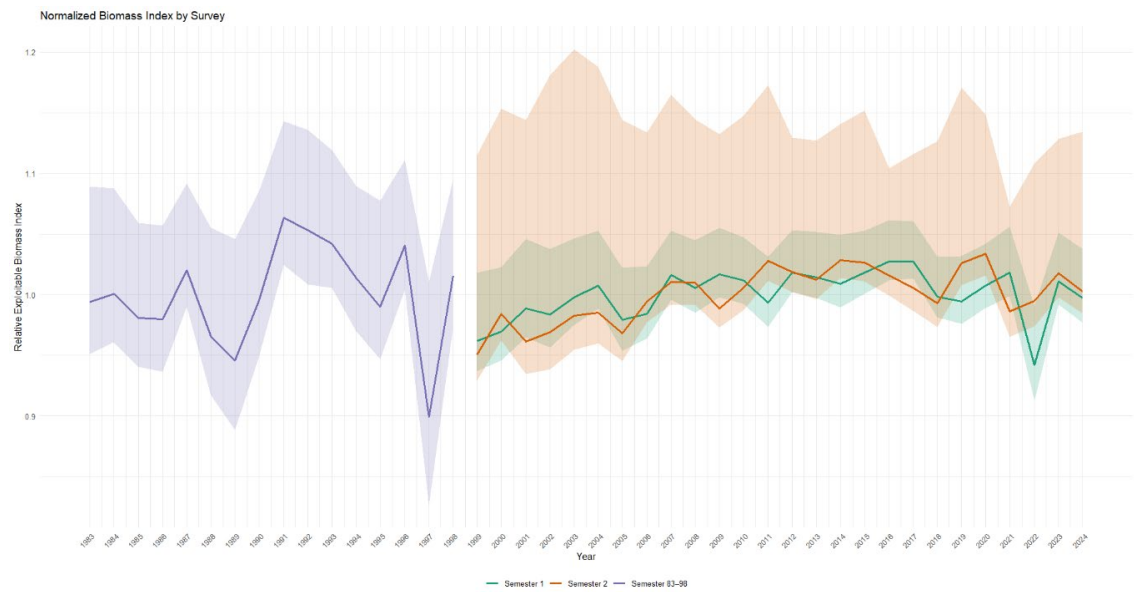


Figure 3.2.12. Brill in Subarea 4 and divisions 3.a and 7.d-e. Standardized exploitable biomass survey index for quarter 1 in Division 3a and Subarea 4 for 1983–1998 (purple line) and for Semester 1 (orange line) and 2 (green line) in Subarea 4 and Division 3a and 7d-e for 1999–2024. The shaded areas represent the 95% confidence interval.

Table 3.2.11. Brill in Subarea 4 and divisions 3.a and 7.d-e. Survey exploitable biomass index for quarter 1 and the coefficient of variation (CV; standardized to the mean) for the period 1983–1998.

Year	Biomass index	CV
1983	513044	1.07
1984	560278.7	0.99
1985	430801.4	0.92
1986	424467.8	0.94
1987	725130.2	0.79
1988	353976	1.07
1989	271308.3	1.22
1990	520670.2	1.06
1991	1283269.6	0.92
1992	1120779.1	0.99
1993	969545.5	0.88
1994	667242.5	0.93
1995	487818.6	1.02
1996	952653.6	0.83
1997	146659.9	1.43
1998	680382.2	0.95

Table 3.2.12. Brill in Subarea 4 and divisions 3.a and 7.d-e. Survey exploitable biomass index for semester 1 (quarter 1) and semester 2 (quarter 3 and 4) and their respective coefficient of variation (CV; standardized to the mean) for the period 1999–present.

Year	Semester 1		Semester 2	
	Biomass index	CV	Biomass index	CV
1999	5468788.3	0.3292912	3561303.6	0.2164768
2000	6098622.4	0.2893551	6007596.4	0.1967412
2001	8477649.3	0.2551454	3862788.4	0.2133402
2002	7381140.1	0.2790067	4071305.3	0.217166
2003	9498508.6	0.2718473	4923220.9	0.1989936
2004	11016182	0.2316336	5322862.5	0.1791243
2005	6965098.4	0.2565263	4363164.2	0.1915186
2006	7570962.1	0.2209402	7198781.5	0.172991
2007	11921248	0.1927103	9531583.8	0.1844077
2008	10092103	0.1952096	9561700.2	0.1687071
2009	12361879	0.1965255	6528438.3	0.1776253
2010	11127216	0.1887932	8607596.8	0.1692314
2011	7795363.5	0.2149751	12144504	0.1475169
2012	12279900	0.1812134	10751738	0.1582169
2013	12308982	0.1961482	9946949.3	0.1550258
2014	11019447	0.2020503	13148133	0.1599844
2015	12773978	0.1878104	12590123	0.1567055
2016	14502806	0.1724411	10331273	0.1608408
2017	15571085	0.1672062	8713640.4	0.173497
2018	9416107.3	0.1939065	7057755.7	0.1792723
2019	9463717.9	0.2018163	12161515	0.1537777
2020	11098546	0.2018644	13538378	0.1568291
2021	12877146	0.2012137	6568383.1	0.1831068
2022	3659091	0.2884518	7761001.5	0.208946
2023	10674524	0.2142745	10832100	0.186636
2024	8843059.6	0.2186945	8682428.2	0.183965

During WGNSSK 2025, the semester 2 biomass index estimated using data from 1999 to 2024 showed unexpectedly high uncertainty (Figure 3.2.12). Exploratory analyses indicated that this was likely driven by limited data coverage during the second half of the year in the western English Channel (Figure 3.2.13 and Figure 3.2.14). To address this, the complexity of the fixed spatial effect in the model—originally estimated separately for each semester—was reduced by applying a single annual two-dimensional smoother instead. This simplification led to a marked reduction in uncertainty around the biomass estimates (Figure 3.2.15). Further sensitivity tests on the assessment model (Figure 3.2.16) confirmed that the simplified index preserved key trends and did not alter the overall perception of the stock. Based on these findings, it was agreed to adopt the simplified index structure going forward. The predictive capacity of the second and third index (indices for semester 1 and 2 between 1999–2024) were analysed through retrospective analysis. For this, the survey indices were recalculated by consecutively removing the last five years of survey information. The survey indices estimates for the first and second semester have acceptable retrospective bias: Mohn’s rho: 0.1 and 0.08 respectively (Figure 3.2.17 and Figure 3.2.18).

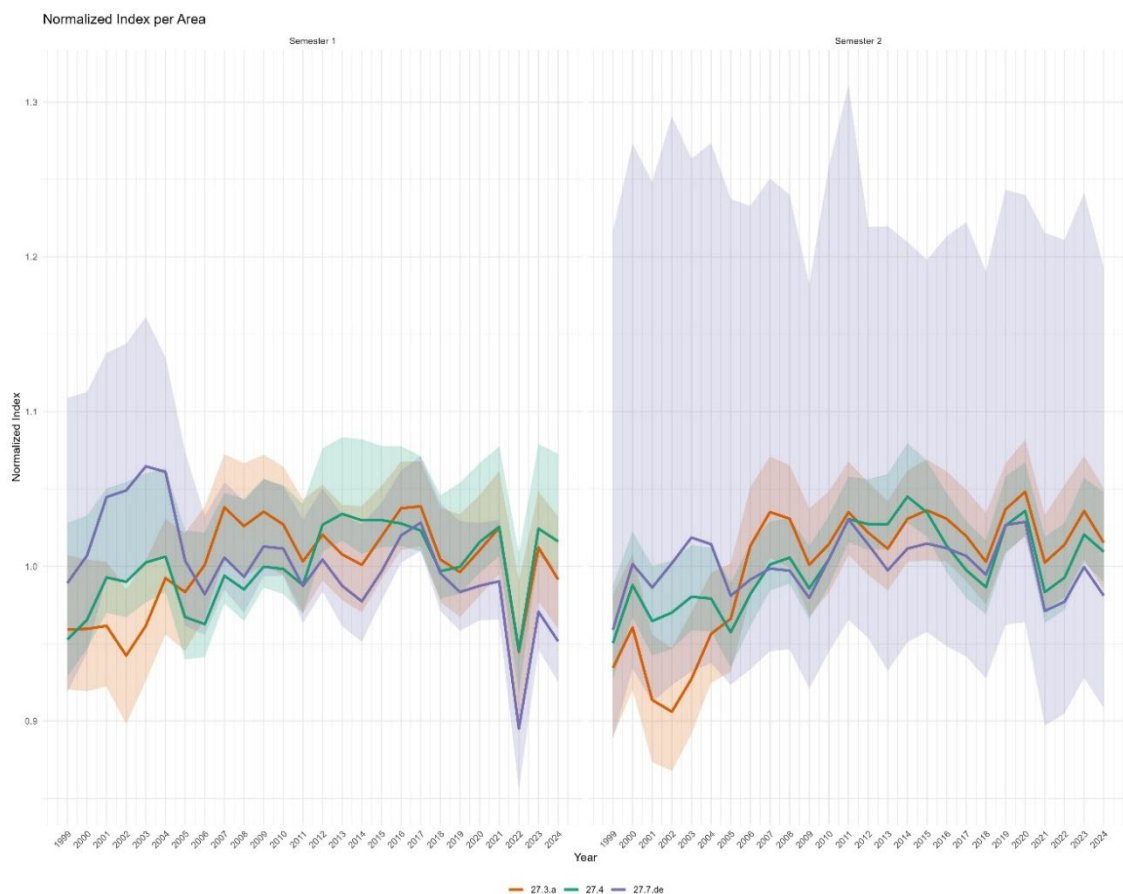


Figure 3.2.13. Brill in Subarea 4 and divisions 3.a and 7.d-e. Exploitable biomass survey index Semester 1 and 2 in Subarea 4 and Division 3a and 7d-e for 1999–2024 from the original survey index model (benchmark). The shaded areas represent the 95% confidence interval.

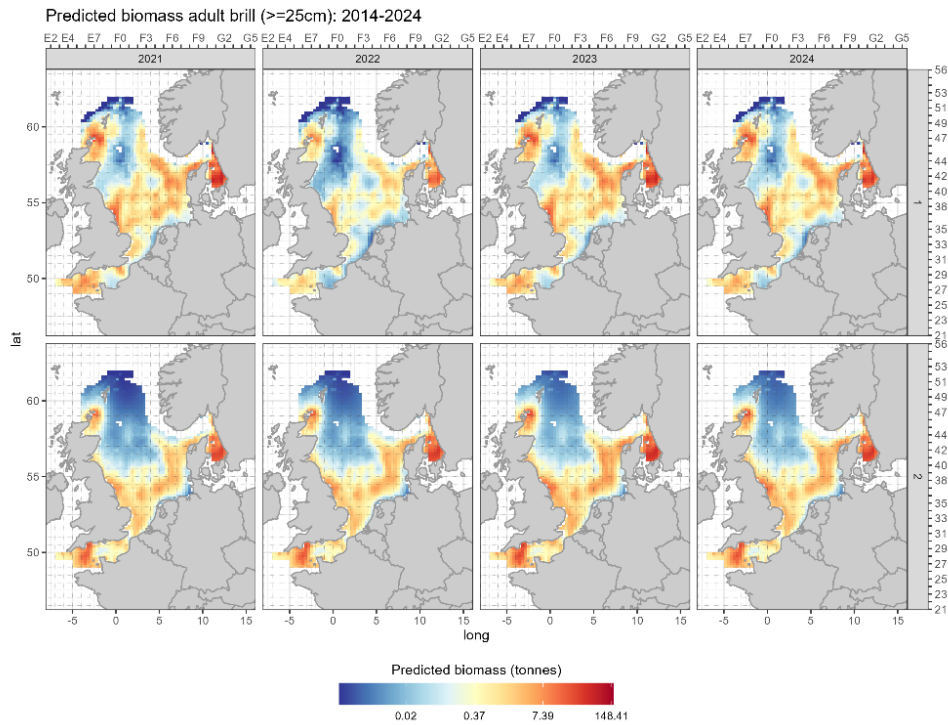


Figure 3.2.14. Brill in Subarea 4 and divisions 3.a and 7.d-e. Predicted exploitable biomass survey index for Semester 1 and 2 in Subarea 4 and Division 3a and 7d-e between 2021–2024.



Figure 3.2.15. Brill in Subarea 4 and divisions 3.a and 7.d-e. Exploitable biomass survey index for Semester 1 (left) and 2 (right) in Subarea 4 and Division 3a and 7d-e for 1999–2024 from the simplified model replacing the fixed spatial effect by semester by a single annual two-dimensional smoother (WGNSSK 2025). The shaded areas represent the 95% confidence interval.

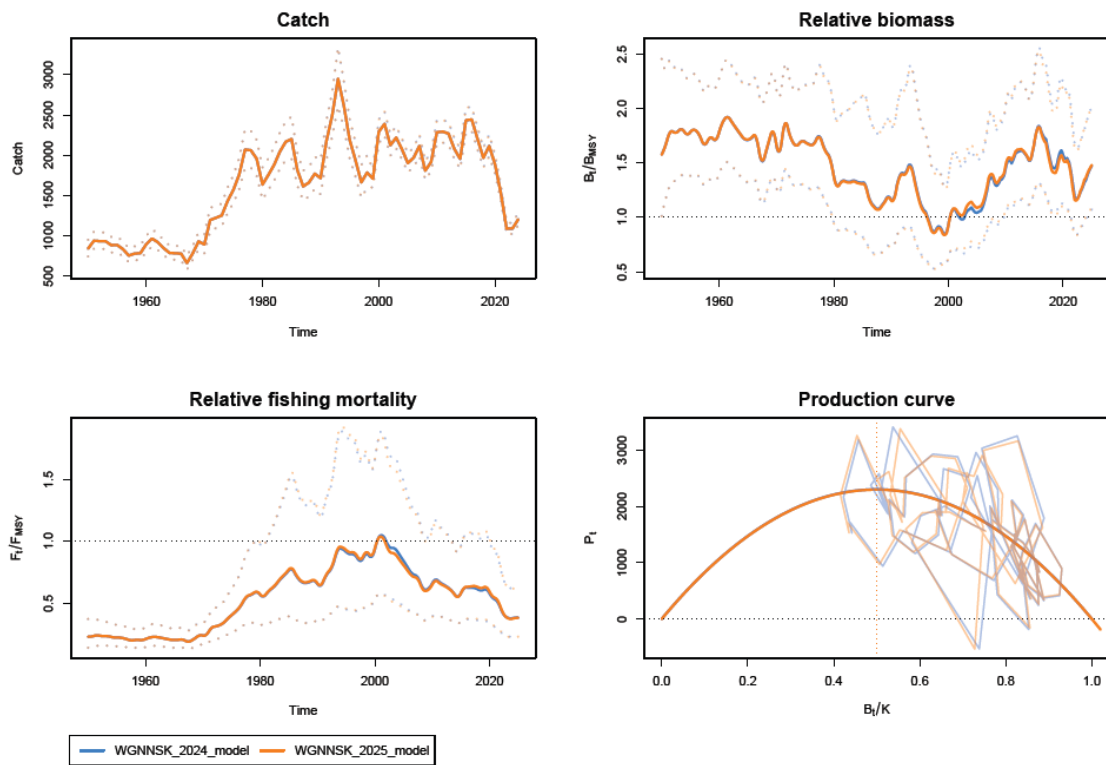


Figure 3.2.16. Brill in Subarea 4 and divisions 3.a and 7.d-e. Sensitivity analysis of the assessment model using the benchmarked approved GAM model (WGNSSK 2024) and the simplified model replacing the fixed spatial effect by semester by a single annual two-dimensional smoother (WGNSSK 2025).

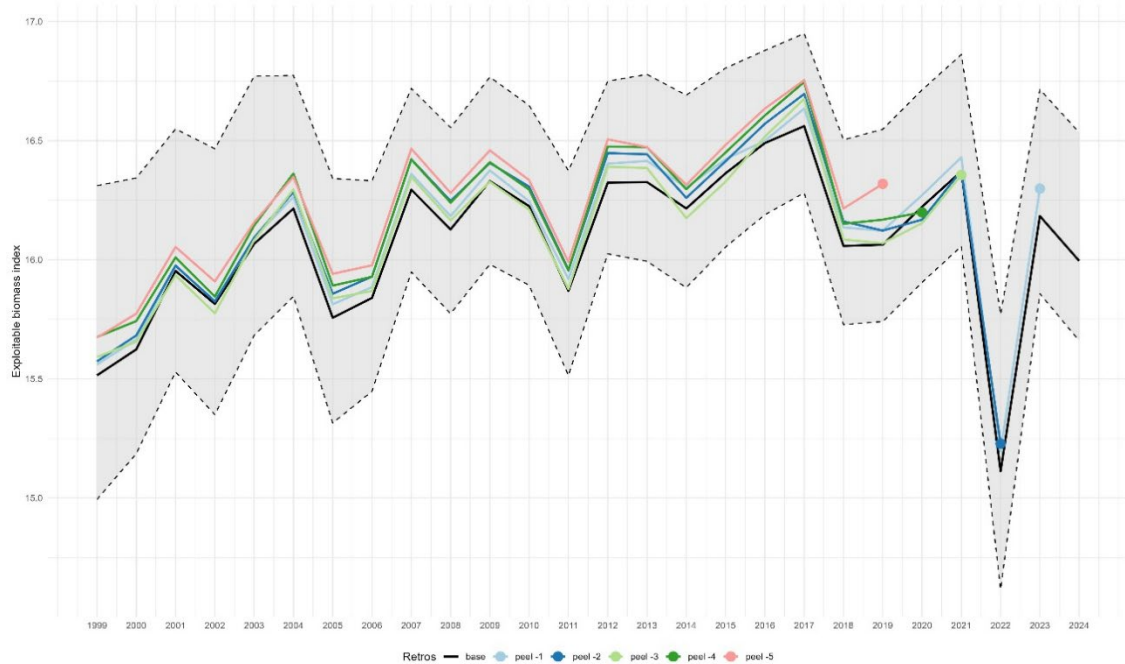


Figure 3.2.17. Brill in Subarea 4 and divisions 3.a and 7.d-e. Retrospective analysis for the exploitable biomass survey indices for Semester 1 in Subarea 4 and Division 3a and 7d-e for 1999–2024 (Mohn's rho = 0.1). The grey shaded areas represent the 95% confidence interval of the base peel.

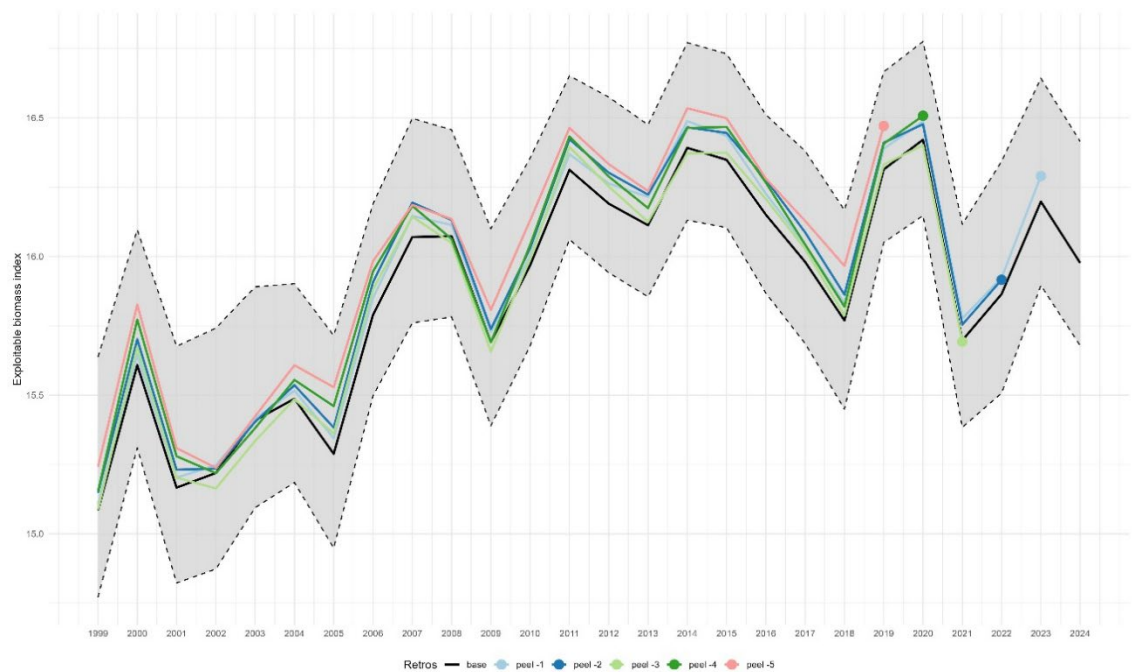


Figure 3.2.18. Brill in Subarea 4 and divisions 3.a and 7.d-e. Retrospective analysis for the exploitable biomass survey indices for Semester 2 in Subarea 4 and Division 3a and 7d-e for 1999–2024 (Mohn’s rho = 0.08). The grey shaded areas represent the 95% confidence interval of the base peel.

3.3 Assessment and short-term forecast

3.3.1 Assessment

The surplus production model in continuous time (SPiCT, Pedersen and Berg, 2017) is used for the assessment of the stock. The *input data* for the stock were 1) a landings series from 1950–2024 with the first part of the time-series from the historical official landings and the most recent part (2014–2024) from ICES InterCatch landings; 2) 3 survey indices: one for semester 1 from 1983–1998, one for semester 1 from 1999–2024 and one for semester 2 from 1999–2024. The survey time-series were trimmed to include only brill equal to or larger than 25 cm in order to obtain a better proxy for the exploitable biomass, which is a prerequisite for a production model, such as SPiCT. The model settings and *priors* are set in accord to the WKBSMYSpiCT 2 analysis (ICES, 2023a) with modification on the *r priors* carried during WGNSSK 2024 (ICES, 2024). The final assessment settings and *priors* for the model parameters as used for WGNSSK 2024 and WGNSSK 2025 are described in Table 3.3.1.

Table 3.3.1. Brill in Subarea 4 and divisions 3.a and 7.d-e. SPiCT settings and input data.

Setting/Data	Values/Source
Catch time-series	Historical landings 1950–2013
	InterCatch landings data 2014–present
Combined survey index semester 1	1999–present, ≥ 25 cm
Combined survey index semester 2	1999–present, ≥ 25 cm

Setting/Data	Values/Source
Combined survey index semester 1	1983–1998, ≥ 25 cm
SPiCT settings	
Standard deviation on the catch (observation) (stdevfacC)	From 1950–1998 stdevfacC = 2; From 1999–present stdevfacC = 1
Standard deviation on the indices (observation) (sdi)	Log(sdi1)~N(log(0.25), 0.3 ²) Log(sdi2)~N(log(0.5), 0.3 ²) Log(sdi3) -> deactivated
Uncertainty ratio of index (observation) to biomass process (alpha)	Deactivated
Uncertainty ratio of catch (observation) to fishing mortality process (beta)	Deactivated
Shape parameter (n)	Schaefer model, Log(n)~N(log(2), 0.01 ²)
Intrinsic growth rate (r)	Log(r)~N(log(0.647), 0.322 ²)
Initial depletion (bkfrac)	log(bkfrac)~N(log(0.8), 0.3 ²)
Discretion time-step (dteuler)	1/16 year (default)

The output of the assessment model is shown in Figure 3.3.1 and Table 3.3.2. Convergence was obtained. The absolute biomass fluctuated around 10 000 tonnes at the beginning of the time-series, but decreased in the 1980s likely as a result of higher fishing pressure (Figure 3.3.5). Fishing pressure increased to unsustainable levels around the year 2000, when F_{MSY} was exceeded (Figure 3.3.1). This caused the biomass to decrease, but nonetheless it remained above $MSY B_{trigger}$. Between 2000 and 2015 when fishing pressure decreased, biomass reacted accordingly and increased again. Landings were rather constant from the 1980s onwards and fluctuated around 2000 tonnes per year. However, landings decreased significantly in the last four years, reaching 1206 tonnes in 2024. The phase or Kobe plot shows that the development of biomass and fishing mortality has been largely sustainable (Figure 3.3.1). In the terminal year of assessment, the relative fishing mortality was below F_{MSY} and the relative biomass was above $MSY B_{trigger}$.

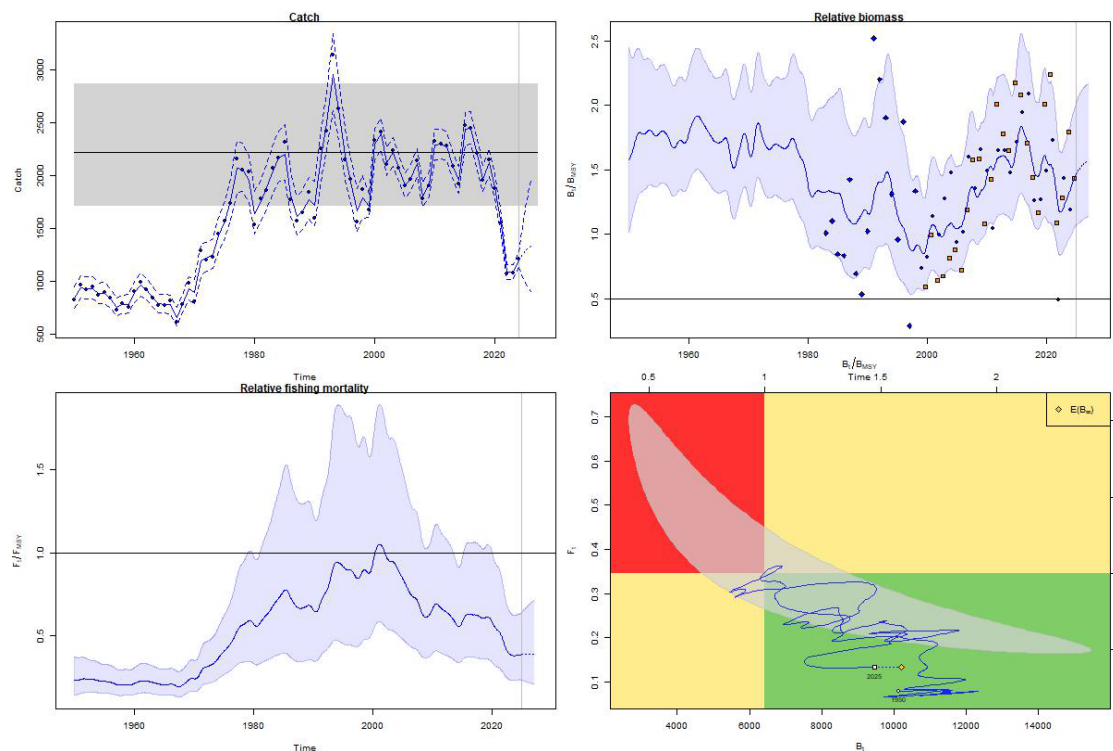


Figure 3.3.1. Brill in Subarea 4 and divisions 3.a and 7.d-e. SPiCT model results from WGNSSK 2025. Top row: Fitted catch and relative SSB. Bottom row: relative F and phase or Kobe plot comparing biomass and F. The grey area in the Kobe plot represents the uncertainty in the relative biomass and F estimates. The dashed lines are 95% CI bounds for absolute estimated values, shaded blue regions are 95% CIs for relative estimates, shaded grey regions are 95% CIs for estimated absolute reference points (horizontal lines).

Table 3.3.2. Brill in Subarea 4 and divisions 3.a and 7.d-e. Assessment results, summary table. The 2025 biomass is the short-term forecast estimate for the intermediate year, assuming that the F process continues unchanged from the last year with observations (F_{sq}).

Year	Relative exploitable biomass			Landings	Discards	Relative fishing pressure		
	Low	B/B _{MSY}	High			Low	F/F _{MSY}	High
1950	1.01	1.58	2.5	827	NA	0.145	0.23	0.37
1951	1.23	1.72	2.4	963	NA	0.155	0.24	0.37
1952	1.32	1.78	2.4	922	NA	0.157	0.24	0.35
1953	1.36	1.80	2.4	947	NA	0.157	0.23	0.35
1954	1.37	1.78	2.3	867	NA	0.153	0.22	0.33
1955	1.38	1.78	2.3	896	NA	0.151	0.22	0.33
1956	1.39	1.79	2.3	842	NA	0.146	0.21	0.31
1957	1.32	1.7	2.2	727	NA	0.139	0.20	0.29
1958	1.32	1.7	2.2	791	NA	0.140	0.20	0.30
1959	1.33	1.71	2.2	758	NA	0.141	0.20	0.30

Year	Relative exploitable biomass			Landings	Discards	Relative fishing pressure		
	Low	B/B _{MSY}	High			Low	F/F _{MSY}	High
				tonnes				
1960	1.39	1.78	2.3	907	NA	0.150	0.22	0.32
1961	1.49	1.90	2.4	987	NA	0.155	0.23	0.33
1962	1.48	1.89	2.4	921	NA	0.153	0.22	0.33
1963	1.42	1.81	2.3	845	NA	0.147	0.21	0.31
1964	1.35	1.74	2.2	776	NA	0.142	0.21	0.30
1965	1.32	1.71	2.2	774	NA	0.141	0.20	0.30
1966	1.34	1.75	2.3	813	NA	0.141	0.20	0.30
1967	1.20	1.60	2.1	611	NA	0.132	0.193	0.28
1968	1.16	1.56	2.1	777	NA	0.145	0.21	0.31
1969	1.34	1.75	2.3	986	NA	0.163	0.24	0.35
1970	1.27	1.66	2.2	810	NA	0.168	0.25	0.36
1971	1.33	1.72	2.2	1289	NA	0.199	0.29	0.43
1972	1.41	1.82	2.4	1205	NA	0.21	0.32	0.47
1973	1.29	1.68	2.2	1226	NA	0.23	0.34	0.51
1974	1.29	1.68	2.2	1450	NA	0.25	0.38	0.58
1975	1.29	1.69	2.2	1568	NA	0.27	0.42	0.65
1976	1.26	1.67	2.2	1735	NA	0.30	0.47	0.74
1977	1.27	1.72	2.3	2159	NA	0.33	0.54	0.88
1978	1.21	1.69	2.4	2047	NA	0.34	0.57	0.96
1979	1.11	1.58	2.3	2036	NA	0.34	0.58	1.00
1980	0.96	1.38	2.0	1538	NA	0.33	0.56	0.97
1981	0.89	1.30	1.91	1775	NA	0.34	0.59	1.04
1982	0.90	1.33	1.98	1861	NA	0.36	0.64	1.14
1983	0.88	1.33	2.0	2064	NA	0.38	0.69	1.27
1984	0.85	1.33	2.1	2161	NA	0.39	0.74	1.41
1985	0.82	1.32	2.1	2316	NA	0.39	0.77	1.51
1986	0.73	1.21	1.98	1774	NA	0.36	0.71	1.41
1987	0.67	1.10	1.81	1571	NA	0.34	0.67	1.31

Year	Relative exploitable biomass			Landings	Discards	Relative fishing pressure		
	Low	B/B _{MSY}	High			Low	F/F _{MSY}	High
				tonnes				
1988	0.66	1.08	1.77	1648	NA	0.35	0.67	1.3
1989	0.72	1.16	1.86	1839	NA	0.35	0.68	1.29
1990	0.73	1.16	1.84	1593	NA	0.35	0.65	1.21
1991	0.83	1.29	2.0	2251	NA	0.39	0.72	1.33
1992	0.91	1.41	2.2	2419	NA	0.42	0.79	1.48
1993	0.93	1.47	2.3	3141	NA	0.47	0.91	1.79
1994	0.83	1.37	2.3	2626	NA	0.46	0.93	1.88
1995	0.68	1.15	1.93	2143	NA	0.45	0.9	1.83
1996	0.62	1.05	1.78	1966	NA	0.44	0.89	1.79
1997	0.54	0.90	1.49	1564	NA	0.44	0.85	1.66
1998	0.57	0.90	1.45	1866	NA	0.47	0.89	1.68
1999	0.56	0.86	1.32	1677	NA	0.49	0.89	1.6
2000	0.63	0.94	1.38	2328	NA	0.56	1.00	1.79
2001	0.72	1.07	1.60	2409	NA	0.58	1.04	1.87
2002	0.67	0.99	1.48	2107	NA	0.54	0.97	1.72
2003	0.70	1.03	1.53	2234	NA	0.53	0.94	1.68
2004	0.74	1.09	1.62	2071	NA	0.5	0.88	1.57
2005	0.71	1.04	1.52	1904	NA	0.46	0.81	1.42
2006	0.77	1.10	1.58	1963	NA	0.43	0.75	1.29
2007	0.93	1.32	1.89	2142	NA	0.41	0.7	1.22
2008	0.94	1.32	1.86	1781	NA	0.37	0.62	1.06
2009	0.96	1.34	1.86	1902	NA	0.38	0.63	1.05
2010	1.05	1.46	2.0	2321	NA	0.40	0.68	1.15
2011	1.08	1.49	2.1	2292	NA	0.40	0.67	1.12
2012	1.15	1.60	2.2	2276	NA	0.38	0.64	1.07
2013	1.16	1.60	2.2	2088	NA	0.36	0.60	1.00
2014	1.11	1.52	2.1	1920	217	0.34	0.56	0.93
2015	1.24	1.69	2.3	2470	207	0.37	0.61	1.02

Year	Relative exploitable biomass			Landings	Discards	Relative fishing pressure		
	Low	B/B _{MSY}	High			Low	F/F _{MSY}	High
	tonnes							
2016	1.31	1.82	2.5	2444	216	0.37	0.63	1.06
2017	1.22	1.72	2.4	2207	210	0.37	0.62	1.06
2018	1.06	1.47	2.0	1956	343	0.36	0.61	1.03
2019	1.07	1.50	2.1	2147	392	0.35	0.61	1.04
2020	1.11	1.57	2.2	1872	182	0.32	0.55	0.95
2021	1.06	1.51	2.1	1553	62	0.29	0.50	0.86
2022	0.85	1.19	1.67	1072	34	0.25	0.41	0.69
2023	0.89	1.23	1.69	1080	82	0.23	0.38	0.63
2024	0.98	1.35	1.85	1206	38	0.23	0.38	0.63
2025	1.07	1.47	2.0					

The prior and posterior distribution show that the estimated parameters were not restricted by the defined prior distributions, as these were defined as vague priors (Figure 3.3.2). The diagnostics of the goodness-of-fit of the model were based on the one-step-ahead residuals (Figure 3.3.3) and process residual (Figure 3.3.4). The OSA residuals showed a significant result in the Shapiro–Wilk test for the first-semester index, indicating a deviation from the assumed distribution. In previous assessments (WGNSSK 2023 and 2024), this test already produced low—but not statistically significant—p-values. The observed deviation is likely driven by an unusually low outlier in the 2022 index, which may itself be the result of reduced sampling during the COVID-19 pandemic. Despite this, the deviation was not considered to undermine the model’s ability to provide reliable information on the stock’s status. However, further investigation is recommended. This should include testing alternative distributional assumptions or model configurations (e.g., a hurdle model) to better handle the low frequency of positive hauls, and evaluating whether the 2022 outlier should be reconstructed or excluded.

The retrospective analysis was done using the recalculated survey index by consecutively removing the last five years of survey information and then fitting the assessment model. The relative fishing mortality and biomass estimates displayed acceptable retrospective bias: Mohn’s rho: -0.078 for B/B_{MSY}, 0.113 for F/F_{MSY} (Figure 3.3.5). The robustness of the model was verified by checking whether the initial values influenced the parameter estimates (Table 3.3.3). Thirty trial runs were performed. Six runs did not converge and the difference in distance was equal to zero, indicating low sensitivity. Finally, a hindcast analysis was carried out to check the ability of the SPiCT model to predict the index by removing a data point and keeping the catch information (Figure 3.3.6). The MASE predictor was calculated and was for both indices below 1, which is considered acceptable.

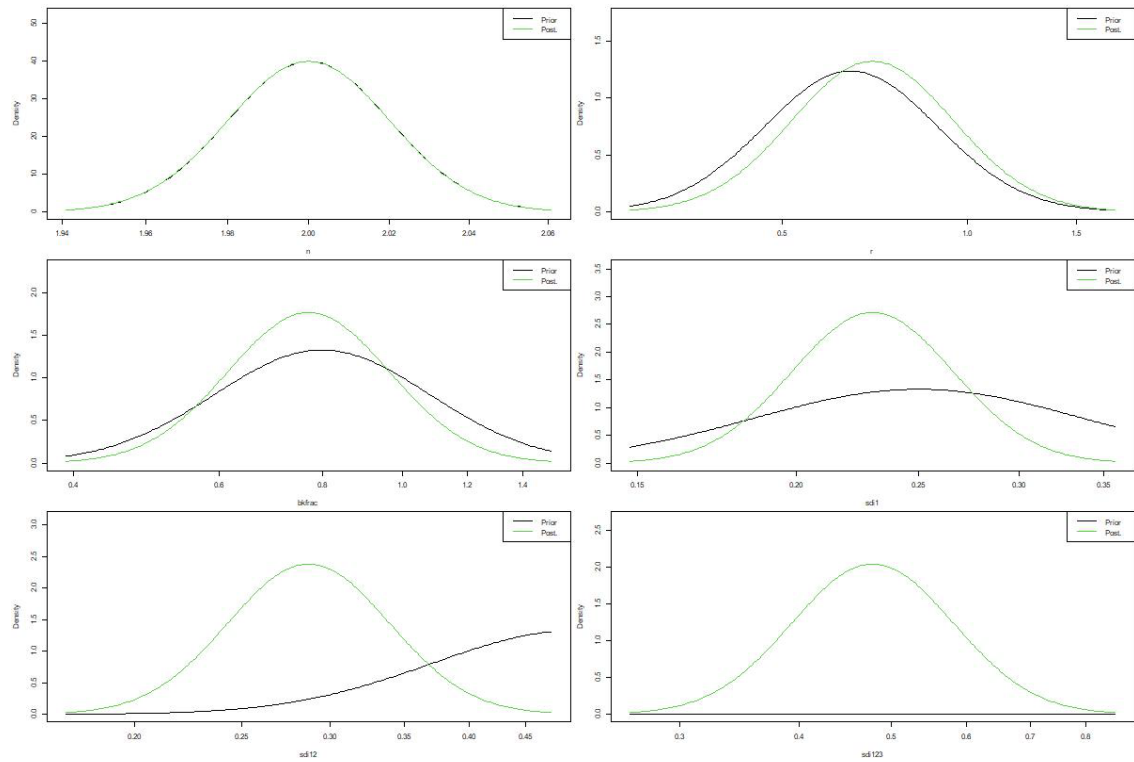


Figure 3.3.2. Brill in Subarea 4 and divisions 3.a and 7.d-e. SPiCT model results from WGNSSK 2025. Prior and posterior distribution of the estimated parameters.

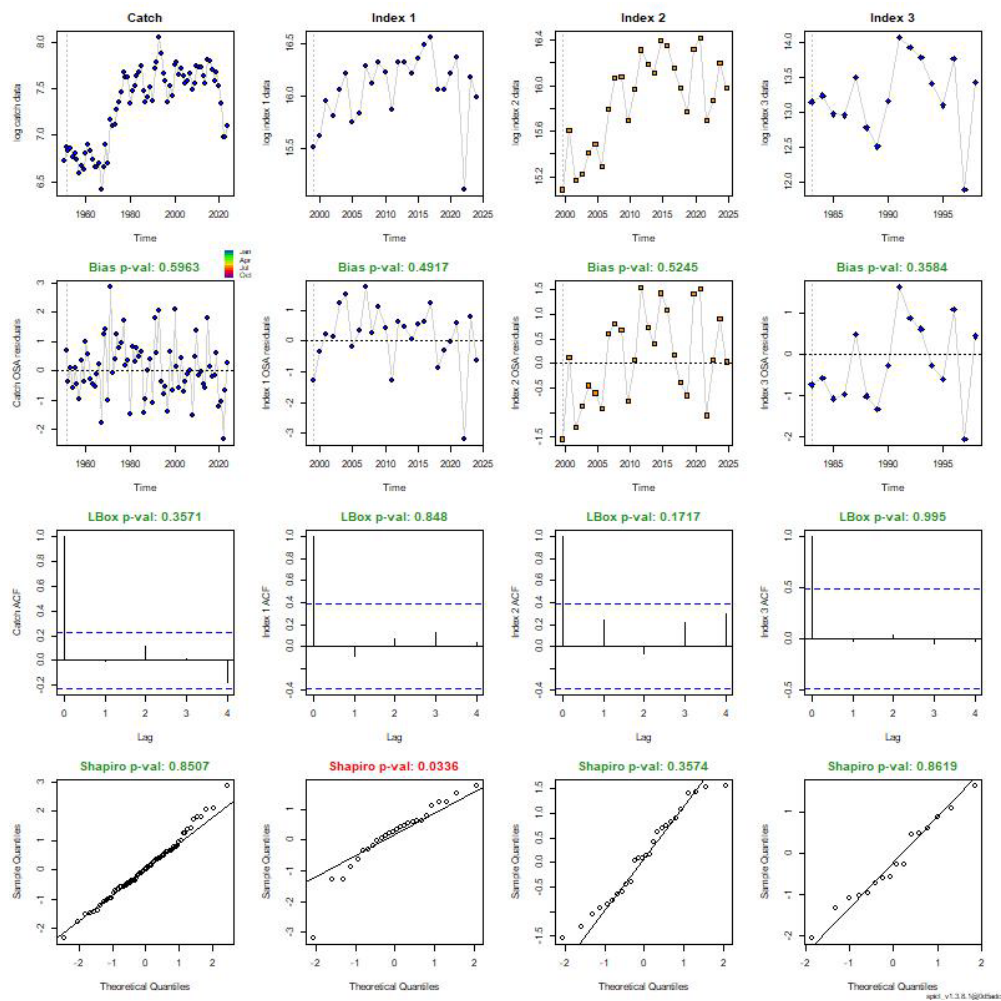


Figure 3.3.3. Brill in Subarea 4 and divisions 3.a and 7.d-e. SPiCT model diagnostics: One-step-ahead residuals.

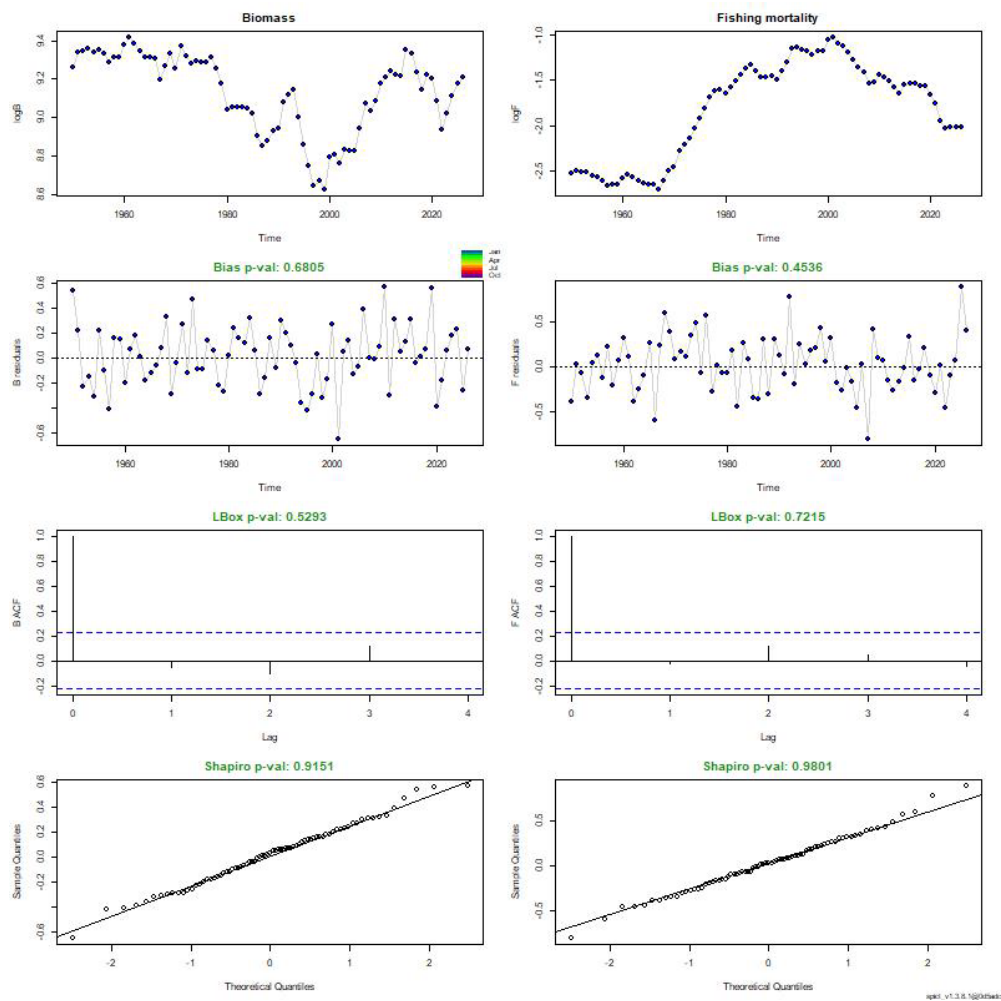


Figure 3.3.4. Brill in Subarea 4 and divisions 3.a and 7.d-e. SPiCT model diagnostics: Process residuals.

Table 3.3.3. Brill in Subarea 4 and divisions 3.a and 7.d-e. Output of the sensitivity analyses concerning the initial parameter values (check.ini). Empty rows represent non-converging runs and are likely caused by a unsuitable combination of initial parameters.

[illegible]

Run	Distance	m	K	q	q	q	n	sdb	sdf	sdi	sdi	sdi	sdci
15	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
16	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
17	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
18	NA												
19	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
20	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
21	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
22	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
23	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
24	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
25	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
26	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
27	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
28	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
29	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03
30	0	2303.23	13142.94	1159.45	940.44	79.26	2	0.15	0.12	0.23	0.29	0.48	0.03

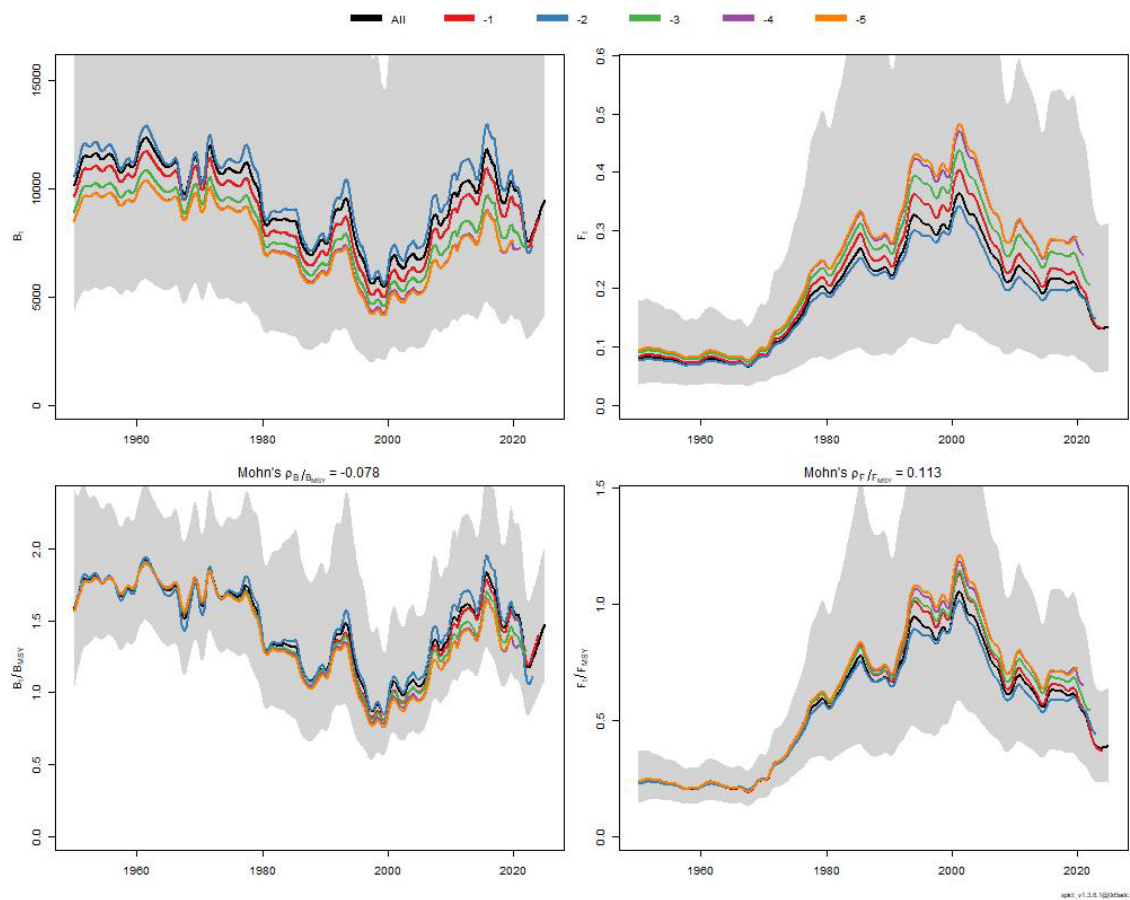


Figure 3.3.5. Brill in Subarea 4 and divisions 3.a and 7.d-e. Retrospective analysis (5 peels) using the recalculated survey index by consecutively removing the last five years of survey information and then fitting the assessment model from WGNSSK 2025. Top row: absolute biomass and absolute F; bottom row: relative biomass and relative F.

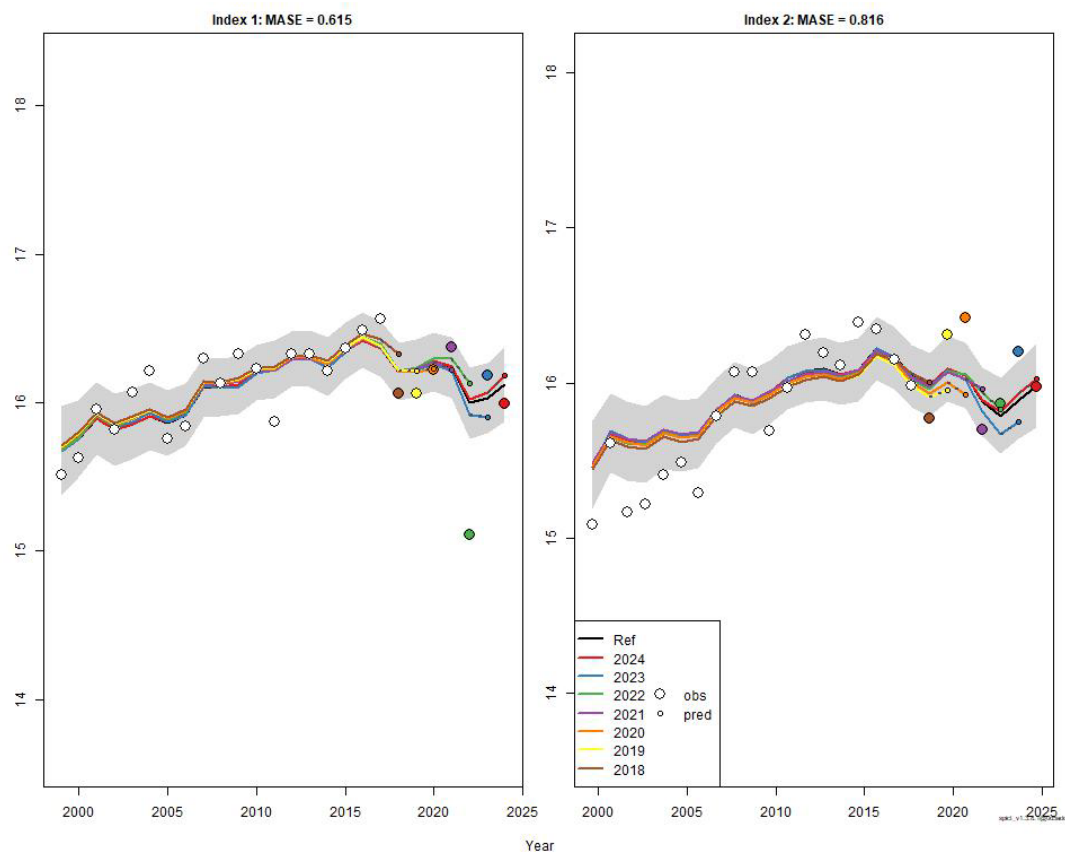


Figure 3.3.6. Brill in Subarea 4 and divisions 3.a and 7.d-e. Hindcast analysis of the SPiCT model from WGNSSK 2025. Left: Index for the first semester between 1999–present; right: Index for the second semester between 1999–present.

3.3.2 Short-term forecast

A short-term forecast is performed using SPiCT. There is no evidence indicating that catches might be constrained by the TAC in 2025. The combined TAC, applicable to both turbot and brill, has been undershot for the last 7 years. Similarly in 2024, only 49.3% of the brill specific TAC was caught in the stock area. Finally, catches clearly decreased since 2020 and remained quite stable in the last 3 years. Consequently, the intermediate year (2025) assumption for the short-term forecast is that the fishing pressure processes continue as observed in the last year, essentially keeping status quo fishing mortality (F_{sq}). An overview of the intermediate year assumptions is shown in Table 3.3.4.

Table 3.3.4. Brill in Subarea 4 and divisions 3.a and 7.d-e. Overview of the intermediate year assumptions.

Variable	Value	Notes
F_{2025}/F_{MSY}	0.39	$F_{sq} = F_{2024}/F_{MSY}$
B_{2026}/B_{MSY}	1.54	Short-term forecast (STF)
Landings (2025)	1286	STF of landings; in tonnes.
Catch (2025)	1345	Based on discard rate (4.4%; average 2022–2024); in tonnes.

3.4 Advice

Advice was calculated for four catch scenarios under the intermediate year assumptions as described in §3.3.2 (F_{sq} ; see Table 3.4.1). Considering that the input data are only landings, the output of the forecast is also a landings advice. The landings advice was then topped up to provide catch advice, considering the average discard rate of the last three years (discard rate 2022–2024 = 4.4%). The $F = 0$ scenario implies no fishing. The $F = F_{sq}$ scenario implies that the F processes continue after the intermediate year ($F = F_{2025}$). F_{MSY} is defined as F/F_{MSY} equal to 1.

Table 3.4.1 Brill in Subarea 4 and divisions 3.a and 7.d-e. Forecast table for the headline advice and alternative catch scenarios. Catch, projected landings and projected discards estimated considering average discard rate 2022–2024 (4.4%). The percent biomass change refers to the biomass in 2027 relative to 2026. The percentage of advice change refers to the catch advice value for 2026 relative to the catch advice value for 2025 (2970 tonnes).

Scenario	Catch advice 2026	Projected landings 2026	Projected discards 2026	F/F_{MSY} (2026)	B/B_{MSY} (2027)	% Advice change
F_{MSY} 35% fractile	3086	2950	136	0.92	1.36	3.9
F_{MSY}	3314	3168	146	1.00	1.33	11.6
$F = F_{sq} = F_{2025}$	1390	1329	61	0.39	1.57	-53
$F = 0$	0	0	0	0	1.74	-100

To provide advice following the MSY approach, the recommendation of WKLIFE X (ICES, 2020) was followed. The basis for the advice assumes fishing mortality $F = F_{MSY}$ during the management year (2026). Then, the advice is the 35th percentile of the projected landings distribution. The use of that percentile instead of the median leads to a more precautionary advice, with no loss of long-term yield and incorporates the uncertainty into the advice. For 2026, this approach leads to a landing advice of 2950 tonnes. Considering the average discard rate of the last three years ($DR_{2022-2024} = 4.4\%$) this resulted in a catch advice of 3086 tonnes. The results for the baseline management scenario and alternatives that are included in the advice sheet are shown in Figure 3.4.1 and Table 3.4.1.

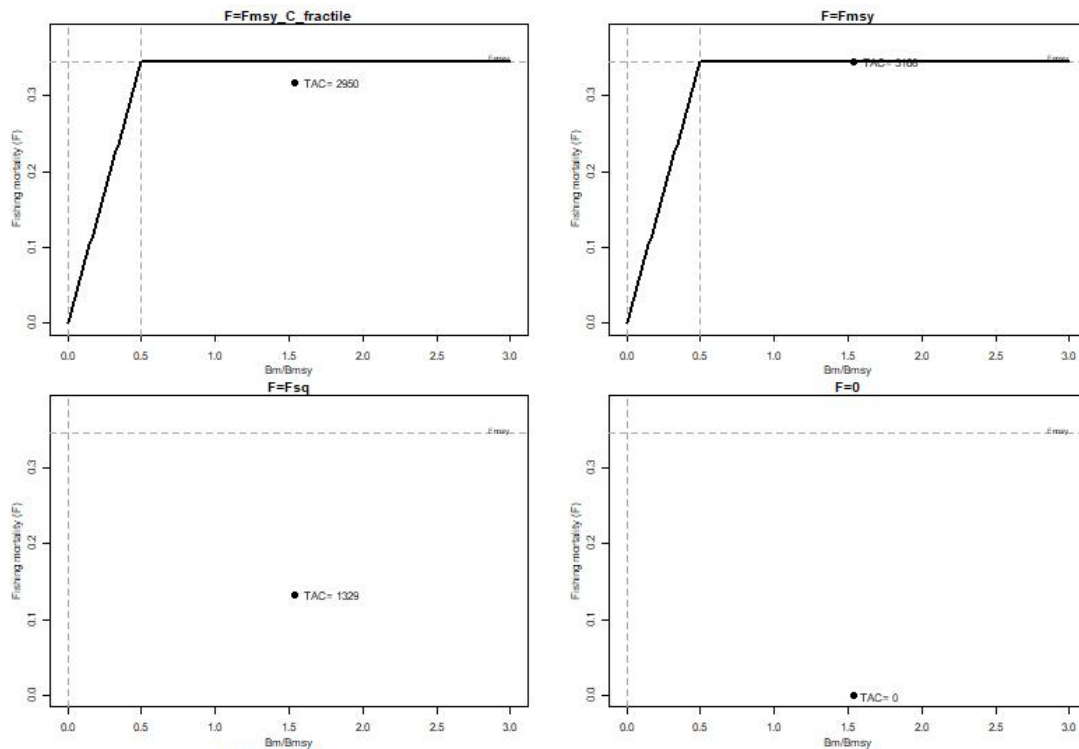


Figure 3.4.1 Brill in Subarea 4 and divisions 3.a and 7.d-e. Management scenarios. The solid line represents the harvest control rule (HCR) applied without incorporating uncertainty (i.e., no fractiles). Under this rule, fishing mortality (F) is set to F_{MSY} when B is above $B_{trigger}$, and decreases linearly to zero as B drops below $B_{trigger}$. Scenarios based on fixed fishing mortalities (e.g., $F = F_{sq}$ or $F = 0$) do not follow a harvest control rule. The vertical dashed line indicates the threshold $B/B_{trigger} = 0.5$. The advice basis, as recommended by WKLIFE X (ICES, 2020), is shown in the top-left corner, which considers the 35th percentile of the projected landings distribution to account for the estimated uncertainty.

The output of the SPiCT assessment suggest that the brill stock is currently in a good state compared to proxy reference points. The change in advice (+3.9%) is mainly due to an increase of the relative exploitable biomass and constant low fishing pressure. (Figure 3.4.2.).

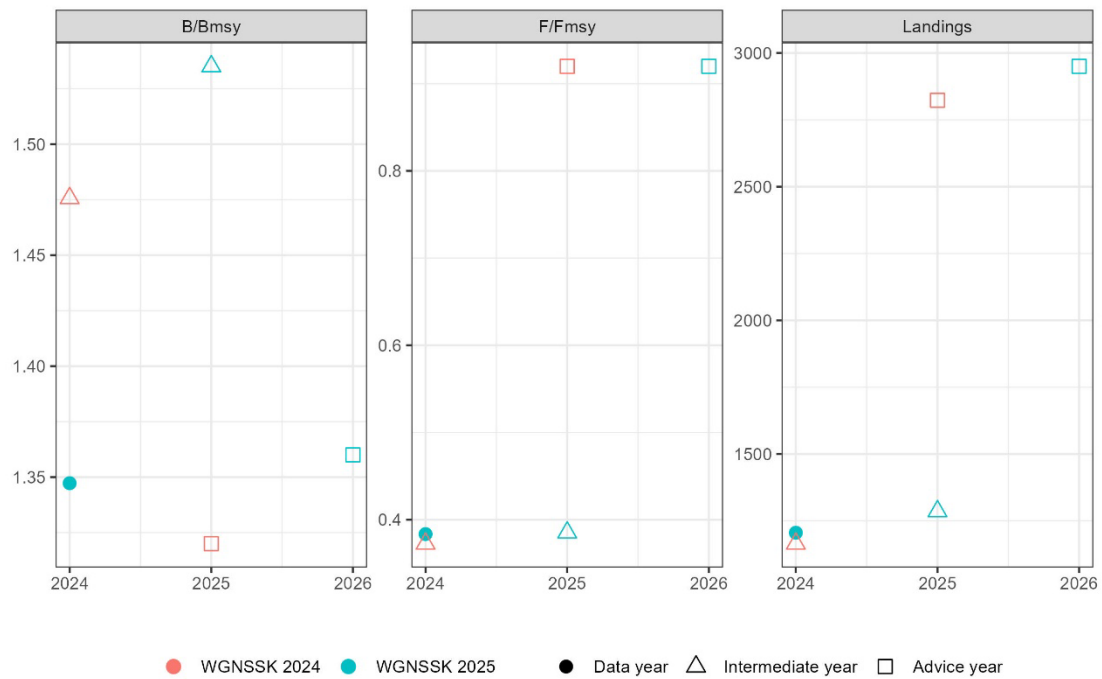


Figure 3.4.2. Brill in Subarea 4 and divisions 3.a and 7.d-e. Change in advice - Comparison between last year’s and this year’s assessment. Note last year’s assessment does not contain the updated survey index calculation and updated data from UK England and the Netherlands.

3.5 Biological reference points

Table 3.5.1 summarizes all known reference points for brill in area 27.3a47de and their technical basis.

Table 3.5.1. Brill in Subarea 4 and divisions 3.a and 7.d–e. Reference points, values, and their technical basis.

Framework	Reference point	Value	Technical basis	Source
MSY approach	MSY B _{trigger}	$B/B_{MSY} = 0.5$	Relative value from the SPiCT model. B _{MSY} is estimated directly from the SPiCT assessment model and changes when the assessment is updated.	ICES (2023b)
	F _{MSY}	$F/F_{MSY} = 1$	Relative value from the SPiCT model. F _{MSY} is estimated directly from the SPiCT assessment model and changes when the assessment is updated.	ICES (2023b)
Precautionary approach	B _{lim}	Not defined		
	B _{pa}	Not defined		
	F _{lim}	Not defined		
	F _{pa}	Not defined		
Management plan	SSB _{mgt}	Not defined		
	F _{mgt}	Not defined		

3.6 Quality of the assessment

- This stock was benchmarked in 2023, leading to a change in assessment method from the *chr* rule (category 3, method 2.2) to SPiCT (category 2; ICES, 2023a, 2023b).
- The previously used commercial LPUE index covering only part of Subarea 27.4 was replaced with three exploitable biomass survey indices covering the whole stock.
- The survey index currently used, assumes a Tweedie distribution. However, considering the low catch rates of brill in survey hauls, a delta log normal (hurdle model) would be more likely to be an ideal model configuration to model the abundance of brill in scientific surveys that are not designed to catch large flatfish such as brill.

3.7 Management considerations

The advice is based on the ICES MSY framework, which recommends the use of the 35th percentile of the projected landing distribution (ICES, 2020). This accounts for the estimated uncertainty. Brill is mainly a bycatch species in fisheries for plaice and sole. As such, fishing mortality for brill is influenced by the management of plaice and sole stocks.

3.8 Benchmark issue list:

Table 3.8.1. Brill in Subarea 4 and divisions 3.a and 7.d–e. Benchmark issue list

Issue	Problem/Aim	Work needed / possible direction of solution	Data needed to be able to do this: are these available / where should these come from?	External expertise needed at benchmark type of expertise / proposed names
(New) data to be considered and/or quantified	Additional M–predator relations	Not at the moment		
	Prey relations	Not at the moment		
	Ecosystem drivers	Not at the moment		
	<i>Other ecosystem parameters that may need to be explored?</i>	Not at the moment		
Stock ID	Explore stock ID and the connectivity with the Baltic	Investigate available data or collect new data to perform genetic analysis	Genetic data or samples from across the stock's distribution area	Experts on stock identity/genetics
New data	Currently, a limited amount of brill data are available in InterCatch. Ask all countries involved in the fisheries to provide available brill data on landings, discards, @age, @length back in time.	Process data in InterCatch.	Data from all countries involved in brill fisheries.	/
Tunning series	Check the potential use of the Dutch industry survey (BSAS).	Analyse data and include in survey index.	Data from the Dutch industry survey available on DATRAS	Dutch experts on the brill-turbot industry survey
	Check the potential use of the new French Survey in 27.7.e	Analyse data and include in survey index.	Data from the French survey to be made available on DATRAS	French experts on the survey
Discards	Discards are not included in the assessment and are lower	Investigate the effect of the inclusion of	Data from all countries involved in brill fisheries.	/

Issue	Problem/Aim	Work needed / possible direction of solution	Data needed to be able to do this: are these available / where should these come from?	External expertise needed at benchmark type of expertise / proposed names
	than 10% in 2024. However, discards could be included if all countries involved in the fisheries provide discard data back in time (before 2014).	discards in the assessment		
Biological Parameters	Recently investigated at the last benchmark in 2023.	/	/	/
Assessment method	Recently investigated at the last benchmark in 2023. However alternative Cat.1 methods could be investigated (e.g. SAM, SS3)	Exploratory runs	Expand the time-series of age/length distribution and discard data available from all countries involved in brill fisheries	/
Biological Reference Points	Recently investigated at the last benchmark in 2023.	/	/	/

3.9 References

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Vandamme, S., 2014. Seascape genetics in support of sustainable fisheries management of flatfish. Doctoral thesis. 304 pp.

3.10 **Supplementary Tables**

3.3.2.1. Brill in Subarea 4 and divisions 3.a and 7.d-e. Official landings (tonnes) of brill in Subdivision 27.3a (Skagerrak/Kattegat) by country, over the period 1950–2024 (Source: ICES Fishstat); *Preliminary.

Year	BE	DE	DK	IE	NL	NO	SE	UK	Total
1950	0	42	234	0	0	0	85	0	361
1951	0	18	260	0	0	4	73	0	355
1952	0	15	170	0	0	1	65	0	251
1953	0	9	175	0	0	0	71	0	255
1954	0	7	155	0	0	1	78	0	241
1955	0	10	150	0	0	0	62	0	222
1956	0	11	163	0	0	0	50	0	224
1957	0	10	110	0	0	0	38	0	158
1958	0	11	166	0	0	0	37	0	214
1959	0	16	175	0	0	0	58	0	249
1960	0	23	272	0	0	0	46	0	341
1961	0	20	255	0	0	0	50	0	325
1962	0	15	207	0	0	0	0	0	222
1963	0	2	120	0	0	0	0	0	122
1964	0	3	106	0	0	0	0	0	109
1965	0	5	155	0	0	0	0	0	160
1966	0	1	187	0	0	0	0	0	188
1967	0	2	106	0	0	0	0	0	108
1968	0	1	100	0	0	0	0	0	101
1969	0	1	99	0	0	0	0	0	100
1970	0	1	97	0	0	0	0	0	98
1971	0	1	104	0	0	0	0	0	105
1972	0	1	120	0	0	0	0	0	121

Year	BE	DE	DK	IE	NL	NO	SE	UK	Total
1973	0	3	131	0	0	0	0	0	134
1974	0	2	200	0	0	0	0	0	202
1975	0	2	167	0	1	0	19	0	189
1976	1	3	185	0	26	0	12	0	227
1977	1	1	276	0	99	0	12	0	389
1978	0	2	178	0	27	0	11	0	218
1979	0	0	156	0	17	0	11	0	184
1980	2	0	69	0	1	0	10	0	82
1981	0	0	54	0	0	0	5	0	59
1982	1	0	64	0	1	0	8	0	74
1983	0	0	73	0	3	0	7	0	83
1984	0	0	89	0	0	0	8	0	97
1985	0	0	100	0	0	0	9	0	109
1986	0	0	94	0	0	0	12	0	106
1987	0	0	93	0	0	0	10	0	103
1988	0	0	91	0	0	0	10	0	101
1989	0	0	88	0	0	0	9	0	97
1990	1	0	116	0	0	0	10	0	127
1991	1	0	81	0	0	7	10	0	99
1992	1	0	123	0	0	7	15	0	146
1993	2	0	184	0	0	10	16	0	212
1994	0	0	191	0	0	12	17	0	220
1995	0	1	124	0	0	13	13	0	151
1996	0	0	94	0	0	12	5	0	111
1997	0	1	83	0	0	11	11	0	106
1998	0	1	108	0	0	10	13	0	132
1999	0	1	126	0	0	13	17	0	157
2000	0	2	112	0	0	12	16	0	142
2001	0	0	73	0	0	13	12	0	98

Year	BE	DE	DK	IE	NL	NO	SE	UK	Total
2002	0	0	66	0	0	12	11	0	89
2003	0	1	99	0	1	12	16	0	129
2004	0	1	119	0	4	15	17	0	156
2005	0	0	101	0	3	16	13	0	133
2006	0	1	105	<0.5	3	16	14	0	140
2007	0	1	119	0	3	15	22	0	160
2008	0	2	138	0	1	13	28	0	181
2009	0	1	98	0	1	14	32	0	146
2010	0	1	95	0	1	9	16	0	122
2011	0	1	103	0	0	15	12	0	131
2012	0	<0.5	89	0	0	16	15	0	120
2013	0	<0.5	70	0	0	9	13	0	92
2014	0	<0.5	59	0	<0.5	8	11	0	79
2015	0	<0.5	104	0	11	8	21	0	145
2016	0	<0.5	125	0	7	8	28	<0.5	168
2017	0	<0.5	131	0	4	8	27	0	170
2018	0	0	90	0	9	9	17	0	125
2019	0	1	93	0	26	3	15	0	139
2020	0	1	112	0	29	3	17	0	162
2021	0	2	102	0	20	3	16	0	143
2022	0	1	57	0	7	2	9	0	76
2023*	0	1	62	0	4	2	10	0	78
2024*	0	1	57	0	9	2	9	0	78

Supplementary Table 3.3.2.2. Brill in Subarea 4 and divisions 3.a and 7.d-e. Official landings (tonnes) of brill in Subarea 27.4 by country, over the period 1950–2024 (Source: ICES Fishstat); *Preliminary. ** French landings of brill in 1999 were provided as a total for all ICES areas (373 t) and not by ICES area.

Year	BE	DE	DK	FR	IE	NL	NO	SE	UK	Total
1950	34	23	39	0	0	108	1	19	183	407
1951	23	19	53	0	0	93	1	19	322	530
1952	21	34	65	0	0	117	3	9	350	599
1953	23	41	49	0	0	130	0	11	376	630
1954	19	37	53	0	0	106	14	7	330	566
1955	23	42	51	0	0	137	3	0	357	613
1956	28	42	47	0	0	156	0	9	276	558
1957	32	31	27	0	0	154	0	8	247	499
1958	43	30	42	0	0	162	0	10	223	510
1959	41	26	30	0	0	125	0	9	219	450
1960	55	28	37	0	0	150	1	8	235	514
1961	102	34	40	0	0	166	0	9	264	615
1962	97	53	42	0	0	214	0	0	238	644
1963	79	52	59	0	0	175	0	0	307	672
1964	79	42	46	0	0	279	0	0	161	607
1965	71	31	56	0	0	281	0	0	127	566
1966	100	26	63	0	0	264	0	0	119	572
1967	138	27	29	0	0	137	0	0	105	436
1968	152	40	43	0	0	274	0	0	110	619
1969	145	38	47	0	0	364	0	0	102	696
1970	114	35	42	0	0	386	0	0	76	653
1971	187	45	72	0	0	720	0	0	94	1118
1972	213	15	65	0	0	665	0	0	51	1009
1973	185	13	55	0	0	710	0	0	39	1002
1974	135	15	68	0	0	905	0	0	44	1167
1975	164	20	76	13	0	925	0	0	44	1242
1976	148	15	65	10	0	940	0	0	45	1223
1977	166	37	88	17	0	1079	0	0	60	1447

Year	BE	DE	DK	FR	IE	NL	NO	SE	UK	Total
1978	175	43	123	26	0	967	0	0	84	1418
1979	188	27	154	10	3	908	0	0	103	1393
1980	129	21	104	8	0	747	0	0	45	1054
1981	148	8	66	5	0	957	0	0	42	1226
1982	182	6	53	11	0	1007	0	0	41	1300
1983	182	7	62	23	0	1153	0	0	28	1455
1984	190	11	73	30	0	1200	0	0	29	1533
1985	187	4	71	35	0	1370	0	0	46	1713
1986	131	5	76	4	0	950	0	0	46	1212
1987	140	5	50	17	0	715	0	0	48	975
1988	102	10	33	18	0	880	0	0	52	1095
1989	112	15	43	9	0	1080	0	0	58	1317
1990	168	30	139	24	0	480	0	0	82	923
1991	205	38	145	28	0	1111	8	0	147	1682
1992	203	59	77	34	0	1196	22	1	218	1810
1993	291	63	118	38	0	1647	14	0	268	2439
1994	208	90	109	28	0	1235	11	0	235	1916
1995	194	67	55	24	0	943	6	0	145	1434
1996	206	47	64	15	0	732	8	0	175	1247
1997	129	48	38	1	0	590	16	0	135	957
1998	160	58	58	11	0	808	16	0	172	1283
1999	161	51	91	NA	0	805	16	0	156	1280
2000	167	77	93	16	0	998	16	0	141	1508
2001	182	66	67	12	0	1075	13	0	158	1573
2002	145	58	52	10	0	907	10	0	120	1302
2003	145	70	57	9	0	934	12	0	119	1346
2004	140	66	77	7	0	772	19	0	168	1249
2005	120	62	89	7	0	716	28	0	138	1160
2006	105	55	75	9	0	765	12	0	154	1175

Year	BE	DE	DK	FR	IE	NL	NO	SE	UK	Total
2007	110	47	52	12	0	854	9	0	156	1239
2008	117	42	86	5	0	650	11	0	93	1004
2009	109	54	96	8	0	786	4	0	105	1162
2010	104	75	97	12	0	1072	4	0	136	1499
2011	101	57	122	13	0	1061	6	0	137	1496
2012	110	71	126	12	0	1084	7	0	122	1532
2013	101	63	123	10	0	972	4	0	118	1390
2014	99	69	96	9	0	857	9	0	117	1255
2015	154	115	122	7	0	1159	1	0	136	1695
2016	175	90	131	8	0	965	1	0	156	1526
2017	138	76	121	7	0	1000	2	<0.5	116	1460
2018	98	80	96	6	0	806	2	<0.5	100	1188
2019	116	132	90	5	0	932	1	0	112	1387
2020	84	99	95	2	0	813	1	<0.5	95	1189
2021	67	59	110	1	0	575	1	<0.5	66	879
2022	31	26	88	1	0	295	1	<0.5	48	490
2023*	43	44	77	<0.5	0	268	2	<0.5	63	497
2024*	62	59	84	<0.5	0	349	2	<0.5	92	649

Supplementary Table 3.3.2.3. Brill in Subarea 4 and divisions 3.a and 7.d-e. Official landings (tonnes) of brill in Subdivisions 27.7.d, e (English Channel) by country, over the period 1950–2024 (Source: ICES Fishstat); *Preliminary. ** French landings of brill in 1999 were provided as a total for all ICES areas (373 t) and not by ICES area.

Year	BE	DE	DK	FR	IE	NL	UK	Total
1950	11	0	0	0	0	0	48	59
1951	8	0	0	0	0	0	70	78
1952	6	0	0	0	0	0	66	72
1953	2	0	0	0	0	0	60	62
1954	1	0	0	0	0	0	59	60
1955	4	0	0	0	0	0	57	61
1956	2	0	0	0	0	0	58	60
1957	4	0	0	0	0	0	66	70
1958	2	0	0	0	0	0	65	67
1959	1	0	0	0	0	0	58	59
1960	6	0	0	0	0	0	46	52
1961	1	0	0	0	0	0	46	47
1962	3	0	0	0	0	0	52	55
1963	1	0	0	0	0	0	50	51
1964	0	0	0	0	0	0	60	60
1965	2	0	0	0	0	0	46	48
1966	0	0	0	0	0	0	53	53
1967	1	0	0	0	0	0	66	67
1968	3	0	0	0	0	0	54	57
1969	2	0	0	121	0	0	67	190
1970	10	0	0	0	0	0	49	59
1971	18	0	0	0	0	0	48	66
1972	20	0	0	0	0	3	52	75
1973	20	0	0	0	0	0	70	90
1974	25	0	0	0	0	0	56	81
1975	24	0	0	55	0	0	58	137
1976	41	0	0	170	0	0	74	285
1977	45	0	0	197	0	0	81	323

Year	BE	DE	DK	FR	IE	NL	UK	Total
1978	58	0	3	227	0	0	123	411
1979	55	0	0	262	0	0	142	459
1980	64	0	2	213	3	0	120	402
1981	83	0	0	271	0	0	136	490
1982	105	0	0	225	0	1	156	487
1983	107	0	0	234	0	1	184	526
1984	114	0	0	226	0	0	191	531
1985	94	0	0	213	0	0	187	494
1986	115	0	0	183	0	0	158	456
1987	126	0	0	216	0	0	151	493
1988	112	0	0	202	0	0	138	452
1989	89	0	0	213	0	0	123	425
1990	99	0	0	249	0	0	195	543
1991	81	0	0	249	0	0	140	470
1992	82	0	0	223	0	0	158	463
1993	78	0	0	256	0	0	156	490
1994	88	0	0	227	0	0	175	490
1995	91	0	0	248	1	0	218	558
1996	105	0	0	240	0	0	263	608
1997	107	0	0	185	1	0	208	501
1998	70	0	0	196	0	2	183	451
1999	97	0	0	NA	0	3	140	240
2000	164	0	0	260	1	4	249	678
2001	212	0	0	256	0	2	268	738
2002	204	0	0	268	0	1	243	716
2003	217	0	0	287	1	1	253	759
2004	165	0	0	259	1	3	238	666
2005	138	0	0	267	0	2	204	611
2006	180	0	0	281	<0.5	3	184	648

Year	BE	DE	DK	FR	IE	NL	UK	Total
2007	205	0	0	325	0	1	212	743
2008	155	0	0	224	0	2	215	595
2009	131	0	0	278	0	1	183	594
2010	145	0	0	340	0	1	213	700
2011	141	0	0	304	0	0	220	665
2012	120	0	0	263	0	1	240	624
2013	142	0	0	238	<0.5	1	224	605
2014	166	0	0	245	<0.5	1	233	645
2015	162	0	0	278	0	2	257	698
2016	143	0	0	286	<0.5	1	290	721
2017	135	0	0	276	<0.5	2	249	663
2018	128	0	0	280	1	2	303	714
2019	103	0	0	287	0	3	267	660
2020	91	0	0	209	0	2	249	551
2021	84	0	0	235	0	3	228	549
2022	76	0	0	230	0	2	203	512
2023*	69	<0.5	0	209	<0.5	2	218	497
2024*	64	<0.5	0	204	0	2	213	484

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